Labor Demand for Digital Skills in Post-2022 Ukraine: Evidence from Online Job Vacancy Data*

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July 19, 2026

Abstract

This paper examines how invasion-related disruptions shape the demand for digital skills. Using a novel dataset of millions of online job vacancies in Ukraine from 2021 to 2025 - classified with large language models and mapped to the ESCO taxonomy - we link vacancy-level skill data to high-frequency ACLED attack intensity measures and estimate time-series models. We document a robust positive association between related fatalities and events and the share of postings requiring digital skills. Our results are robust across different time-series specifications and alternative attack intensity measures. This shift is driven by both within- and between-occupational shifts and by remote work adoption as a mediating channel. The rising digital skill share partly reflects the relative resilience of digital hiring against contracting total vacancy volumes rather than

*We thank Jooble for providing the dataset underlying this research and Yuliia Antoniuk for her collaboration. We thank Bo Pieter Johannes Andree, Alexandra Avdeenko, Oleksandr Faryna, Niels Planel, Abla Safir, Oleksandr Talavera, Nithin Umapathi for their valuable comments and suggestions. We thank participants of the Economic Modelling 2025 Conference "Ukraine: Economic Insights for Future Policy Actions", Trinity College Dublin, August 25-26, 2025; seminar participants at the National Bank of Ukraine Research Seminar, November 2025; and participants of the research conference on "Harnessing web data for next-generation skills intelligence", 28-29 May 2026, organized by CEDEFOP, for valuable comments and suggestions. We also thank two anonymous reviewers for their revisions and suggestions. This publication uses the ESCO classification of the European Commission.

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absolute digital expansion. These findings suggest that digitalization reshapes labor demand in ways that carry lasting implications for workforce planning and reconstruction policy.

Keywords: Labor Demand, Technological Change, Large Datasets, National Security, War, Conflict

JEL Codes: J23, O33, C55, H56

1 Introduction

Armed conflict exerts profound and multifaceted effects on labor markets. It contracts the available workforce through conscription and displacement, while simultaneously heightening the risks associated with physical tasks, leading to acute shortages in both labor supply (Collier and Duponchel 2013) and shifts in the demand for skills (Pham, Talavera, and Wu 2023). Although suggestive evidence indicates that employers alter their skill requirements during wartime, a systematic examination of the mechanisms driving such shifts remains lacking (Pham, Talavera, and Wu 2023) and little is known about skills demand during wartime.

This paper investigates whether digitalization can mitigate labor shortages and structural disruptions induced by war, invasion, and conflict. We posit that, faced with a reduced labor pool and damaged infrastructure, firms accelerate the adoption of digital technologies, automate routine processes, and reorganize tasks to maintain productivity and operational resilience. As a result, firms signal demand for digitally intensive skills. International evidence shows that war, invasion, and conflict lead to the destruction of factories, machinery, and transportation networks, consequently eroding capital stocks (Chauvin and Rohner 2009). Moreover, wars lead to changes in existing institutions and market functioning, raising transaction costs and operational uncertainties (Verwimp, Justino, and Brück 2019). In such an environment, organizations must seek novel operational modalities and technological innovations to survive and adapt (Udomsaph 2023). Evidence of other shocks, such as the COVID-19 pandemic (Forsythe et al. 2020) or recessions (Hershbein and Kahn 2018), have shown how severe disruptions can cause significant changes in skills demand. Closely related, Modestino, Shoag, and Ballance (2016) and Modestino, Shoag, and Ballance (2020) show that recessions induce systematic upskilling in vacancy postings — firms raise stated skill requirements when labor markets are slack — and that this process partially reverses as conditions tighten.

We study this question in the context of Russia's full-scale invasion of Ukraine. Conscription and emigration have significantly reduced the domestic labor supply and firms face challenges to fill vacancies (World Bank 2024). In 2023, nearly two out of 10 businesses who had closed reported a lack of workers as the main reason (12 percent in 2024) (World Bank 2024). In addition, the ongoing situation increases the hazards of on-site work due to the security risk, damaged assets, and increased uncertainty (World Bank 2023; World Bank 2024). Roles requiring on-site attendance—such as plant operator positions—have consistently been hard to fill across all regions in Ukraine (World Bank 2024).

Both the private and public sector are increasingly digitalizing their services. For example, in December 2023, the Ukrainian government announced e-services for those who have lost their homes since 2022 (EU4Digital 2024), online marriage registration services (initially targeted to military personnel) (EAPL 2025), and in July 2024 it launched a one-stop-shop for starting and managing a business online (EGA 2024). The government also provides several e-services targeting IDPs (launched in May 2023) and veterans (launched in July 2025) (Kyiv Global Government Technology Center 2025a). Businesses are accelerating digitalization in response to the current situation, too. A survey on the state of business digitalization in Ukraine conducted in 2024 revealed that six out of 10 surveyed companies have accelerated digitalization (Kyiv Global Government Technology Center 2025b). Both remote work and cost optimization are frequent reasons for this trend among interviewed companies.

To investigate this question, we leverage online job postings collected from an online job vacancy aggregator, the Jooble platform, from January, 2021 to June, 2025. Jooble, a prominent job vacancy aggregator, compiles listings from various Ukrainian job portals, providing a rich dataset for labor market analysis.[1] Utilizing large language models (LLMs), we extract and classify information from online job postings to identify demanded occupations, and required skills. We then map the resulting occupations and skills to the official ESCO classification. Using the official ESCO classification, we calculate the relative demand

[1]Jooble is a global product-based IT company active in 67 countries and has been operating in Ukraine for over 18 years.

for digital skills in the Ukrainian labor market. We combine this data with information on the number of attack-related events and fatalities provided by ACLED. We then run time-series models to investigate the relationship between the current situation in Ukraine and the relative demand for digital skills.

Our results reveal a positive correlation between post-2022-related events and fatalities and the demand for digital skills, indicating that war, invasion, and conflict could shift the labor market toward digitally intensive tasks and approaches. The positive association also persists after controlling for seasonal patterns through time fixed effects, broader labor-market conditions captured by weekly vacancy volumes, and the technological shocks induced by the rollout of ChatGPT and a second AI wave. The results are further robust to alternative time-series specifications, including moving-average smoothing of both the outcome and the regressor, as well as an Almon polynomial lag model that flexibly captures dynamic effects.

Economic theory provides a robust foundation for our hypothesis that wartime pressures accelerate digitalization and reshape labor demand. Capital–labor substitution models predict that scarce labor supplies will prompt firms to substitute toward capital (Solow 1956), including software, robotics, and cloud-based solutions. Task-based frameworks suggest that routine tasks are most susceptible to automation, non-routine cognitive and interactive tasks complement digital tools, and entirely new tasks emerge alongside technological progress (Autor, Levy, and Murnane 2003; Autor 2015; Acemoglu and Restrepo 2018).

Empirical evidence confirms that ICT adoption systematically raises the task content of employment toward non-routine cognitive activities across industries and countries (Michaels, Natraj, and Van Reenen 2014), and that automation and digitalization polarize labor demand by hollowing out middle-skill routine occupations while expanding demand at both ends of the skill distribution (Gregory, Salomons, and Zierahn 2022). This implies that the rise in digital skill demand should operate through two reinforcing channels: a within-occupation channel, whereby existing job profiles are augmented with digital requirements as routine tasks are automated and non-routine cognitive tasks complemented by digital tools; and a between-occupation channel, whereby demand shifts toward inherently more digital-intensive occupations. Furthermore, search-and-matching models indicate that prolonged job vacancies and elevated recruitment costs incentivize investments in automation and digital platforms to bridge workforce gaps (Yashiv 2007; Diamond 2011).

Crucially, these adjustments are not instantaneous: as attack intensity rises, the barriers to on-site work increase and labor supply disruptions accumulate through displacement and conscription, so the effect of attack-induced disruption on stated skill demand is expected to emerge with a lag and accumulate gradually over several weeks rather than materializing contemporaneously — a dynamic we account for through an Almon polynomial lag specification. While our empirical strategy identifies robust associations between attack intensity and digital skill demand, we caution throughout that our estimates are associative rather than causal.

Remote work adoption emerges as a mediating channel linking attack-related disruption to higher digital-skill requirements: as barriers to on-site work rise, firms shift toward remote-compatible roles that are inherently more digital-skill-intensive. This mechanism is consistent with evidence that a substantial share of jobs in modern economies can be performed remotely (Dingel and Neiman 2020), and that remote work adoption - once triggered by a large aggregate shock - tends to persist well beyond the initial disruption (Barrero, Bloom, and Davis 2021). We further show that the rising digital skill share partly reflects the relative resilience of digital hiring against a backdrop of sharply contracting total vacancy volumes, rather than an absolute expansion of digital hiring activity — a distinction with direct implications for how the results should inform workforce policy. Lastly, Oaxa-Blinder decompositions reveal that the results are related to shifting demand towards higher-skilled occupations with a larger concentration of digital skills as well as an increasing demand for digital skills within existing occupations.

By integrating theoretical perspectives with empirical analysis, this paper contributes to a growing literature on conflict, vacancies, and skill demand by providing large-scale vacancy-based evidence on how attack-related disruptions reshape the demand for digital competencies. Our contribution is distinctive in four respects: (i) the scale of the dataset and LLM-based ESCO classification pipeline; (ii) the combination of high-frequency ACLED conflict intensity data with vacancy-level skill data in a time-series framework; (iii) the focus on a full-scale invasion as a distinct and understudied shock relative to the economic disruptions examined in prior work; and (iv) the application of an Oaxaca-Blinder within/between occupation decomposition to skill demand during periods of absence of peace.

Our study talks to the literature established by Acemoglu and Autor (2011), who discuss the implications of skills, tasks, and technologies on employment and earnings. It is most closely related to a newly emerging literature studying the interaction between demographic change and automation (Acemoglu and Restrepo 2022) and a limited set of studies analyzing the impact of war, invasion, and conflict on innovation and technological change (Anosike 2018; Gross and Sampat 2023; Maltsev 2023; Clemens and Rogers 2023). Studies that analyze the impact of automation and artificial intelligence on labor demand, as explored by Acemoglu and Restrepo (2019), Acemoglu and Restrepo (2020), and Acemoglu et al. (2021), are also related to our analysis. For example, Eloundou et al. (2024) show that LLMs have substantial exposure potential across a wide range of occupations, with higher-skill cognitive roles disproportionately affected — a finding that motivates our inclusion of a ChatGPT rollout control to isolate the absence-of-peace-driven component of digital skill demand from concurrent AI adoption trends.

More broadly, our work contributes to a rich literature documenting that war and violence leave persistent spatial and human capital scars — through physical destruction, landmine contamination, disrupted migration, and reshaped financial behavior — while also generating unexpected social and institutional transformations (Blattman and Miguel 2010; Bauer et al. 2016; Blumenstock et al. 2024; Chiovelli, Michalopoulos, and Papaioannou 2025; Prem, Purroy, and Vargas 2025; Riaño and Valencia Caicedo 2024; Jung, Lim, and Park 2025). Our contribution to this strand is to show that war, conflict, and invasions can also reshape the skill composition of labor demand, with potentially durable consequences for reconstruction recovery trajectories.

Importantly, the developments documented in this paper occur against a backdrop of pre-existing structural pressures. Akcigit et al. (2025) show that Ukraine's economy was already experiencing declining business dynamism and rising incumbent concentration prior to 2022. Firm-level survey evidence from World Bank (2023) and World Bank (2024) corroborates our vacancy-based results, showing that Ukrainian firms are actively restructuring operations in response to post-2022-related pressures - lending credibility to the labor demand shifts we document at scale.

Our research offers valuable insights into the dynamic nature of skills demand in Ukraine, informing policymakers, educators, and industry stakeholders in their efforts to develop responsive workforce development strategies. It closes an important gap in labor market intelligence in Ukraine. Accurate labor market information is crucial for efficient labor markets by providing accurate, timely, and accessible data. While we study a specific situation, the mechanisms we identify likely generalize to other war, invasion, conflict settings, offering actionable lessons for reconstruction workforce planning. The findings carry direct implications for workforce policy during times of hostilities, post-conflict reconstruction planning, and the design of digital skills investment programs in fragile and conflict-affected states more broadly.

2 Data

2.1 Data Extraction and Processing

We construct a large-scale database capturing skills demand in Ukraine, drawing on data provided by Jooble.[2]

[2]Jooble is an IT company founded in 2006 that operates Jooble.org, an international job search engine active in 67 countries. The platform functions as an online job vacancy aggregator, using a proprietary

In Ukraine, Jooble collects job postings from 3,490 sources—2,570 directly from employers and 920 from other aggregated platforms. Ukrainian employers can register directly on Jooble, enabling the platform to comprehensively cover their job-related websites. The resulting database includes classified job portals, corporate websites, and other platforms featuring job listings. Jooble has been active in the Ukrainian market for 18 years, providing a robust and longitudinal web archive. Its web crawler scans the entire internet every 24 hours to ensure the data remains current. The final dataset provided by Jooble is in JSON format and includes information on the job title, job description, region of the job opening, salary details, number of clicks, and the dates the vacancy was created and expired.

We process the full Jooble database from January, 2021, to June, 2025, a total set of 17.1 million vacancies. We first clean the dataset by removing duplicates, removing unnecessary text from job titles and vacancy descriptions, and then standardize the language in the job descriptions. We also conduct a language detection exercise given that the original Jooble data for Ukraine comes in multiple languages (Ukrainian, Russian, Czech, Polish, and English). The multilingualism of the job posts is related to the fact that the Jooble website in Ukraine also captures remote jobs, as well as jobs abroad, such as from the Czech Republic, Poland, Germany, as well as several other European countries (Greece, Portugal, Slovakia, and Lithuania). Appendix A provides more details on the data cleaning.

Data Processing Overview. We construct the final dataset through a six-stage processing pipeline. Jooble provides the raw job vacancy postings in JSON format. As a first step of the data management, we apply cleaning, deduplication, and language filtering. We extract candidate skills from vacancy texts by applying multilingual sentence embeddings. We then match them to the ESCO skill taxonomy based on semantic similarity. In the third stage, vacancies are assigned a provisional ESCO/ISCO-aligned occupation code using a large language model under structured output constraints. This provisional assignment is not treated

algorithm and keyword-based search to scrape Google for job postings across a given country. It also incorporates automated mechanisms to filter out spam, phishing, and inappropriate job listings.

as the final occupation entry. Instead, it is used as an anchor for the fourth stage, where each vacancy is resolved to a terminal ESCO occupation entry and enriched with the official ESCO skill bundle linked to that occupation. The revised verification step combines exact and fuzzy matching on ESCO preferred and alternative labels with a code-based fallback, `esco_code_similarity`, which uses the provisional occupation code and hierarchical ESCO taxonomy search to recover the closest valid terminal entry. After this five-step verification cascade, only 0.08% of postings remain unclassified. We standardize and harmonize geographic information. Lastly, we aggregate the processed data into weekly and monthly panels and combine it with ACLED measures of related events and fatalities. A detailed technical description of each processing stage is provided in Appendix A.

Appendix A also provides a worked example tracing a representative vacancy through the pipeline, from raw input to final ESCO occupation and skill assignment.

Modularity and Extensibility. The pipeline is a modular and extensible framework. While the present paper focuses on empirical analysis, the underlying architecture separates data ingestion, skill extraction, classification, enrichment, and aggregation into configurable stages. This design allows future extensions—such as alternative taxonomies, updated models, or additional outcome measures—to be incorporated through metadata or configuration changes without restructuring the core workflow.

2.2 Skills and Occupation Extraction

This subsection details the skill- and occupation-specific components of the processing pipeline introduced above. We first identify candidate skill requirements by embedding vacancy texts with the *paraphrase-multilingual-mpnet-base-v2* model from *Sentence-Transformers* (Reimers and Gurevych 2019). The model produces multilingual sentence embeddings suitable for semantic similarity matching across Ukrainian, Russian, Polish, Czech, and English vacancy texts.

We compare these vacancy-text embeddings with embeddings of official ESCO skill labels, using both preferred and alternative labels. In the candidate skill extraction step, we apply a cosine similarity threshold of 0.5 to prioritize recall and retain up to 25 candidate skills per vacancy. These candidate skills, together with the job title, form the structured input for the occupation-classification stage. Where at least five candidate skills are available, the vacancy is classified using *gpt-4o-mini* under a fixed prompt and structured JSON output schema; where fewer skills are available, the model receives the job title and a truncated vacancy description of up to 1,000 characters. Low-confidence cases are reprocessed with *gpt-4o*.

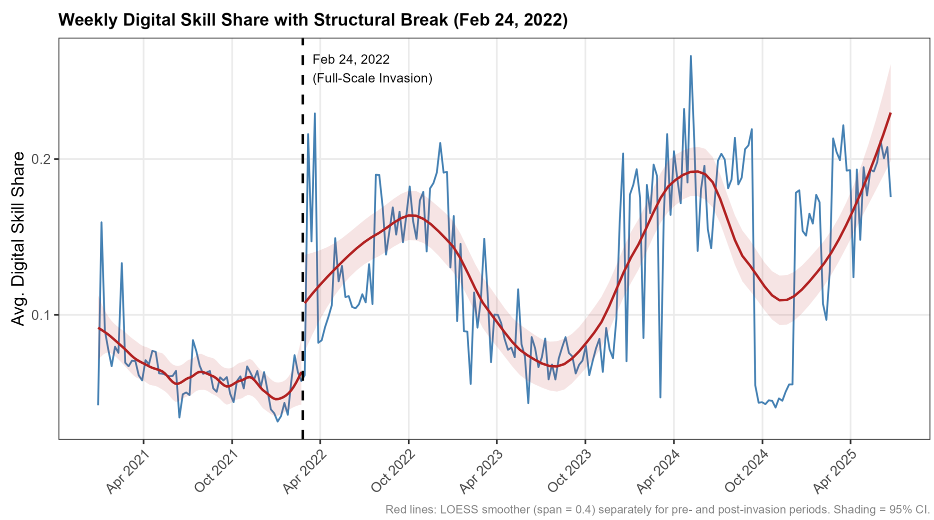
The LLM output is treated only as a provisional ESCO/ISCO-aligned occupation anchor. Final occupation assignment is obtained through the ESCO verification and enrichment stage described in Appendix A, where postings are resolved to verified terminal ESCO occupation entries and enriched with the official ESCO skill bundle linked to that occupation. Appendix A provides more details on the skills and occupation extraction process.

To assess the accuracy of the occupation-classification step, we conduct a manual validation exercise on a stratified random sample of 200 records drawn from the April 2024 dataset. The validation evaluates classification accuracy across successive levels of the ESCO hierarchy, since occupation assignment determines the terminal ESCO entry from which the associated skill bundle is retrieved. Classification accuracy reaches 91.0% at the major-group level and 88.0% at the sub-major-group level, indicating reliable broad occupational placement, and 80.0% at the four-digit unit-group level, which is directly relevant for skill enrichment. Full validation results and threshold-sensitivity checks for the Stage 4 matching procedure are reported in Appendix A.

Occupation classification is used as an intermediate step to identify the verified terminal ESCO occupation entry and retrieve its associated ESCO skill bundle. We then construct the outcome by counting all digital ESCO skills (1,201 skills), as identified by the official ESCO classification, and dividing this number by the total number of ESCO skills observed in a given period, such as a week or month. This relative measure captures the digital intensity of skill demand and reduces sensitivity to changes in total scraping volume.

Figure 1 reveals that the relative demand for digital skills has increased over the period January, 2021, to June, 2025. The post-2022 trend exhibits a pronounced cyclical pattern, with the digital skill share peaking in autumn and troughing in spring of each year. This seasonality likely reflects recurring labor market rhythms — such as hiring cycles and budget planning — and is absorbed in the empirical analysis by calendar-quarter fixed effects, which we include in all specifications.

Figure 1: Share of digital skills demand in all skills based on Jooble data (Jan, 2021–June, 2025)



Professional roles demonstrate the greatest concentration of digital skills, followed by managerial roles (Table 1). Figure B2 reveals that the relative demand for professionals has increased over time, similarly to the demand for Managers. This is in line with findings from (Pham, Talavera, and Wu 2023) showing an increase in the demand for high-skilled occupations post-2022. The occupational distribution in Table 1 is in line with information from household survey data collected by the World Bank between 2023-2025, showing that Professionals are the biggest occupational group in Ukraine, followed by Managers (Table B1).

Table 1: Average digital skill intensity by occupation group (percent)

Occupation group	Number of vacancies	Digital skill share
Professionals	780392	16.80
Managers	79786	5.12
Technicians and associate professionals	296714	4.08
Clerical support workers	121906	2.48
Craft and related trades workers	286659	1.83
Plant and machine operators and assemblers	185749	1.30
Service and sales workers	264449	0.71
Elementary occupations	188105	0.68
Skilled agricultural, forestry and fishery workers	7537	0.20

2.3 Data Limitations

Importantly, the data may not fully reflect the true structure of labor demand across Ukraine. Since Jooble aggregates postings primarily from online sources, occupations that are typically filled through informal networks, in-person recruitment, or public employment centers—such as public sector jobs, rural work, or highly specialized roles—may be underrepresented. Additionally, disparities in internet access, especially in targeted regions, may further bias the coverage of job ads, limiting the representativeness of the data across all oblasts and sectors.

Several limitations of the data construction procedure merit explicit acknowledgement. First, the LLM-based occupation-classification step is subject to classification error. Manual validation on a stratified sample of 200 job postings shows an error rate of 9.0% at the one-digit major-group level and 20.0% at the four-digit unit-group level, which is the level the model was instructed to target. Appendix A provides the corresponding validation details. This matters because the digital skill indicator is constructed from ESCO skill bundles linked to the terminal occupation entries assigned to each vacancy. Fine-grained occupation-level errors may therefore introduce noise into the skill-enrichment stage. However, because the outcome is aggregated to weekly and monthly shares, such measurement error is expected primarily to attenuate estimates rather than create systematic bias, unless classification errors are correlated with attack intensity. We consider such systematic correlation unlikely, as all pipeline parameters were fixed prior to sample construction and do not vary across regions or time periods.

Second, the translation of ESCO skill labels into Russian may introduce noise for rare or highly specialized skill labels because ESCO does not provide an official ESCO taxonomy in Russian language. This is mitigated by semantic similarity matching rather than exact keyword matching, which reduces sensitivity to minor wording imprecision in translated labels.

Third, only 0.08% of vacancies remain unclassified after all five steps of the ESCO verification cascade in Stage 4; these records are excluded from the analysis.

A further limitation is that observed trends in online vacancy data may partly reflect fluctuations in scraping activity, source availability, platform architecture, and reposting behaviour rather than genuine shifts in labour demand. This concern is well documented in the online-vacancy literature, where changes in scraping coverage, deduplication rules, and multi-platform reposting can create time-varying discrepancies between online posting counts and underlying vacancy stocks (Turrell et al. 2019; Napierala, Kvetan, and Branka 2022; Kurekova, Beblavy, and Thum-Thysen 2015). The issue is also relevant in the Ukrainian context, where multi-portal reposting and repeated publication of unfilled positions generate substantial duplication in raw vacancy collections (Sarioglo and Cymbal 2020). We address this concern in three ways. First, the pipeline implements multi-step deduplication combining identifier-based checks with content-based matching of job titles, descriptions, salary ranges, and regions. Second, observations are aggregated to weekly and monthly panels, reducing sensitivity to idiosyncratic daily scraping fluctuations. Third, and most importantly, the main outcome is the share of digital skills among all extracted ESCO skills rather than the raw number of digital-skill vacancies. This relative measure is less sensitive to changes in total scraping volume and is consistent with established practice in studies using online vacancy data to measure skill and task composition (Pham, Talavera, and Wu 2023; Safir and Muller 2019).

Notwithstanding these limitations, the vacancy-based results in this paper find corroboration in independent firm-level survey evidence. The World Bank's *Firms through the War* reports (World Bank 2023; World Bank 2024) document that Ukrainian firms are actively restructuring operations in response to post-2022-related pressures, with nearly two out of ten businesses that closed in 2023 citing a lack of workers as the primary reason, and a majority of surviving firms reporting accelerated digitalization. A 2024 survey on the state of business digitalization in Ukraine found that six out of ten surveyed companies had accelerated digitalization since 2022, with remote work adoption and cost optimization cited as the primary drivers (Kyiv Global Government Technology Center 2025b).

Overall, while useful, the Jooble data should be interpreted with caution and ideally complemented by other sources to gain a more comprehensive view of skill demand in Ukraine.

2.4 ACLED data

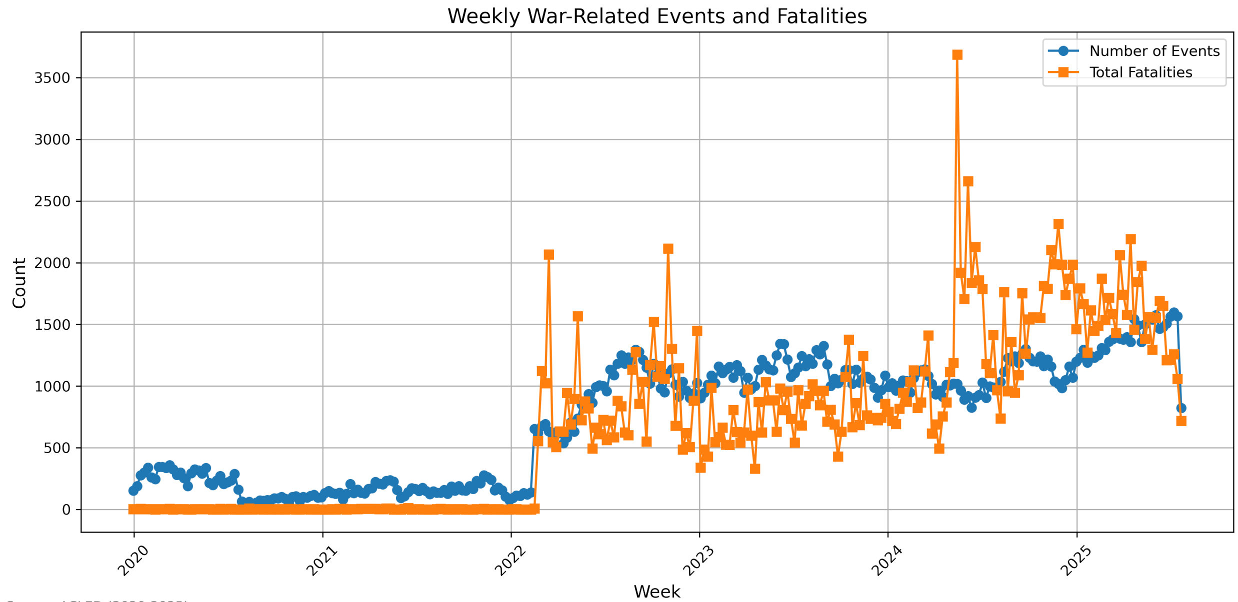
To analyze how the current situation has affected the demand for digital skills in Ukraine, we rely on data from the Armed Conflict Location and Event Data Project (ACLED) (ACLED 2023). ACLED offers a high-resolution, event-level dataset capturing instances of political violence, protests, and strategic developments across more than 200 countries and territories. Each record represents a discrete incident with standardized fields for event type (e.g., battle, riot, demonstration), date, precise location (down to city or village), and actors involved (including state forces, rebel groups, militias, and civilians). The dataset also records reported fatalities, enabling quantitative assessment of attack intensity and human impact over time. ACLED compiles data through a multi-stage process combining automated media scraping, human coding, and cross-validation by regional experts.[3]

Figure 2 depicts the total weekly number of related events and fatalities in Ukraine. The

[3]Source materials include local and international news outlets, government statements, and verified social media reports, which are triangulated to maximize accuracy and minimize reporting bias. Rigorous quality controls detect duplicate or false reports and harmonize actor nomenclature across geographies.

figure clearly reveals how the post-2022 situation resulted in a significant increase in the number of violent events and fatalities in Ukraine.

Figure 2: Total weekly number of post-2022-related events and fatalities in Ukraine (2020 - 2025)



Source: ACLED (2020-2025).

2.5 Other Data Sources

We use additional data sources to generate information on potential confounding factors.

We use two complementary series from the Ukrainian energy monitoring platform (energy-map.info): (i) scheduled electricity outage hours per week (hourly observations, January 2023 – September 2025), and (ii) the weekly count of electricity consumers disconnected specifically due to combat actions, sourced from the Ukrainian Ministry of Energy (March 2022 - March 2025). The Ministry of Energy series is preferable for identification because it covers nearly the full sample period and isolates the attack-caused component of infrastructure disruption rather than aggregate outage intensity. Both variables are set to zero for the pre-2022 period. Moreover, both variables are standardised to zero mean and unit variance to allow comparison of coefficient magnitudes across controls.

3 Methodology

To test the common directional prediction generated by the theoretical frameworks outlined in the introduction - that attack-induced disruption is associated with a positive shift in digital skill demand - we estimate the following reduced-form regression:

$$\mathrm{Y}_t \;=\; \beta_0 \;+\; \beta_1 \,\log(\text{fatalities})_t \;+\; \sum_{m=1}^{11} \delta_m \,\mathrm{D}_{m,t} \;+\; \varepsilon_t \tag{1}$$

where

- Y_t is the average share of digital skills in period t,
- $\log(\text{fatalities})_t$ is $\log(1 + \text{weekly ACLED fatalities})$ in period t,
- δ_m captures calendar-quarter or calendar-month fixed effects,
- β_0 is the intercept, β_1 measures the semi-elasticity of the digital skill share with respect to attack intensity, and
- ε_t is an error term.

$D_{m,t} = 1$ if time period t falls in calendar quarter or month m (with one period omitted as the reference category) and 0 otherwise. These dummy variables absorb any recurring seasonal or calendar-month shocks - holidays, fiscal cycles, training programs - so that β_1 reflects the within-quarter/month relationship between fatalities and the digital skill share. By effectively comparing each time period to itself across years, this specification purges calendar-driven noise and sharpens the precision and validity of our key coefficient.

We run regressions at the weekly and monthly level. We estimate our main models at the weekly frequency, which captures high-frequency adjustments in hiring in response to shocks induced by the current situation in Ukraine. Weekly variation is economically meaningful in Ukraine, where missile strikes, power outages, and front-line movements occur on a weekly rhythm. To ensure that results are not driven by short-term volatility, we re-estimate the specifications at the monthly level.

To account for the highly skewed distribution of attack intensity and to obtain economically meaningful elasticities, we model fatalities (and events) in logarithms. Weekly attack indicators exhibit substantial variation and frequent extreme spikes, making a linear specification inappropriate and highly sensitive to outliers. The log transformation compresses these extreme values, stabilizes variance, and improves the linear fit between attack intensity and labor-demand responses. Substantively, the log specification captures the idea of diminishing marginal effects - an increase from 0 to 10 events is likely to have a much larger impact on employer behavior than an increase from 100 to 110.

A natural concern with our baseline specification is that attack intensity, measured by $\log(1 + \text{fatalities})$, may be correlated with regional economic conditions, internet penetration, platform usage, and sectoral composition, all of which could independently shift the digital skill share of job postings. We address this concern by subsequently adding additional controls: the vacancy count, military skill share, an indicator for the ChatGPT rollout, and an indicator for the second AI-wave rollout.

Throughout, we report heteroskedasticity- and autocorrelation-consistent (HAC) standard errors following Newey and West (1987), using a Newey-West bandwidth of four lags, which spans approximately one calendar quarter at the weekly frequency.

All specifications include calendar-quarter fixed effects (δ_q, $q \in \{\text{Q1, Q2, Q3, Q4}\}$) to absorb within-year seasonal variation in online labour-market activity. Figure B3 plots the mean digital skill share and mean log fatalities by calendar quarter for each year of the sample, revealing systematic within-year variation in the outcome across years. Formal tests confirm this pattern statistically. A non-parametric Kruskal-Wallis test rejects the null hypothesis that the distribution of the digital skill share is equal across the four calendar quarters ($H(3) = 9.95$, $p = 0.019$; Table B2). The ACF of the raw digital skill share series (Figure B4) shows persistent positive autocorrelation at lags corresponding to quarterly multiples ($k \cdot 13$ weeks), a characteristic signature of quarterly periodicity. After removing the within-calendar-year component via quarter demeaning, these peaks attenuate markedly (Figure B5), confirming that the fixed effects absorb the seasonal signal. The Friedman test does not reject consistent year-over-year repetition of the seasonal profile ($\chi^2(3) = 0.60$, $p = 0.896$), reflecting that attack-driven seasonal patterns in Ukraine are not rigidly tied to a fixed calendar cycle - which motivates the use of flexible quarter indicators rather than a parametric seasonal decomposition.

3.1 Additional Specifications

To assess the robustness of the relationship between attack intensity and the demand for digital skills, we estimate three further classes of specification: a Interrupted Time Series (ITS) design, moving-average smoothing, a polynomial distributed-lag (Almon) model, and a set of time-series robustness checks addressing spurious regression.

Moving-average smoothing. Because weekly fatality counts exhibit considerable short-term volatility, we replace the raw log(1 + fatalities) regressor with 4-, 8-, and 12-week backward moving averages and re-estimate the baseline model using these smoother inputs. This approach reduces the influence of idiosyncratic weekly spikes and allows us to evaluate whether the estimated relationship is driven by short-run noise or reflects broader medium-run dynamics. Formally, for window width $w \in \{4, 8, 12\}$ and each lag $k \in \{0, 1, 2, 3\}$ we estimate a separate OLS model:

$$y_t \;=\; \alpha + \beta_k \, \bar{x}^{(w)}_{t-k} + \boldsymbol{\gamma}' \mathbf{z}_t + \varepsilon_t, \tag{2}$$

where $\bar{x}^{(w)}_{t-k}$ is the w-week backward moving average of log(1 + fatalities) ending at week $t-k$, and $\mathbf{z}_t$ contains quarter fixed effects and, in the fully-controlled variant, vacancy count, military skill share, a linear time trend, and the ChatGPT and second-AI-wave indicators.

Polynomial distributed-lag (Almon) model. While moving averages reveal whether the result is robust to variance reduction in the attack measure, they do not identify how the timing of attack shocks propagates into employer behaviour. To examine the dynamic response more explicitly, we estimate a polynomial distributed-lag (PDL) model following Almon (1965), which retains the full weekly lag structure but constrains adjacent lag coefficients to lie on a smooth polynomial curve. This approach mitigates multicollinearity across highly correlated weekly lags, improves statistical precision, and yields an interpretable lag profile showing how the effect builds, peaks, and fades over time. The specification is:

$$\mathrm{Y}_t \;=\; \alpha_0 \;+\; \sum_{k=0}^{K} \beta_k \,\log(\text{fatalities})_{t-k} \;+\; \sum_{m=1}^{11} \delta_m \,\mathrm{D}_{m,t} \;+\; \varepsilon_t \tag{3}$$

where K is the maximum lag length (we use $K = 8$ weeks, spanning approximately two months) and the lag coefficients $\{\beta_k\}_{k=0}^{K}$ are constrained to lie on a polynomial of degree d (we set $d = 3$):

$$\beta_k \;=\; \sum_{j=0}^{d} \gamma_j \, k^j, \quad k = 0, 1, \ldots, K. \tag{4}$$

We argue that firms need time to revise job postings, retrain HR processes, and adjust skill requirements. An 8-week window captures a plausible full adjustment cycle without extending so far that the lags pick up unrelated medium-run trends. Moreover, our polynomial choice is motivated by the fact that with K=7, a degree-2 polynomial imposes a strictly symmetric (parabolic) lag profile — it cannot accommodate an asymmetric response where the effect ramps up, peaks, then tapers at different speeds. A cubic allows this asymmetry, meaningfully reducing multicollinearity relative to the unrestricted distributed lag.

Substituting (4) into (3) transforms the model into a regression on $d + 1$ constructed regressors $Z_{j,t} = \sum_{k=0}^{K} k^j \log(\text{fatalities})_{t-k}$, from which the γ_j are estimated by OLS. The original lag coefficients β_k are then recovered via (4). The polynomial constraint reduces multicollinearity across adjacent weekly lags, improves precision, and yields a smooth dynamic response function. The cumulative (long-run) effect of a unit change in log(fatalities) is $\sum_{k=0}^{K} \beta_k$. This approach is well-suited to our setting, where the effect of attack intensity is expected to evolve gradually rather than discontinuously week by week.

Interrupted Time Series. We complement the continuous-regressor baseline with an ITS design (Box and Tiao 1975) that allows both the intercept and the slope of the digital skill share series to shift discretely at the date of 24 February 2022. Before choosing between the continuous and structural-break representations, we test formally for parameter stability using three complementary procedures: a Chow test at the known break date, a CUSUM test on OLS residuals, and an Andrews Sup-F test at an unknown break date (Appendix Table B5) (Andrews 1993). All three reject parameter stability at conventional significance levels, with 24 February 2022 confirmed as the maximising break date, motivating the ITS as an alternative to the continuous log(fatalities) regressor. The ITS design has the additional advantage of being robust to the mechanical collinearity between a persistently rising attack measure and a linear time trend, which attenuates the continuous regressor in fully-controlled specifications.

We estimate the ITS as follows:

$$Y_t = \alpha + \beta_1 t + \beta_2 Post_t + \beta_3 (t - t^*) \cdot Post_t + \gamma_q + \varepsilon_t, \tag{5}$$

where Y_t is the weekly share of vacancies requiring digital skills, t is a linear time trend, $Post_t$ is an indicator equal to one for weeks on or after the respective date (February 24, 2022), t^* denotes the week so that $(t - t^*) \cdot Post_t$ equals zero pre-2022 and counts weeks elapsed since the 24th of February thereafter, and γ_q are calendar-quarter fixed effects. The coefficient β_2 captures the immediate level shift at the break date, while β_3 captures the change in the weekly trend slope after this date.

Spurious regression checks. A further concern is that both the digital skill share and log fatalities may be integrated of order one, I(1), raising the possibility that the estimated relationship is spurious (Granger and Newbold 1974). We address this concern through four complementary steps reported in Appendix Tables B7-B10: establishing the order of integration via ADF (Dickey and Fuller 1979), PP (Phillips and Perron 1988), and KPSS (Kwiatkowski et al. 1992) tests (Table B7); testing for cointegration via the Engle-Granger two-step procedure (Engle and Granger 1987) (Table B8); estimating an Error Correction Model (ECM) to validate the long-run relationship (Engle and Granger 1987) (Table B9); and checking robustness under the trend-stationary alternative by residualising both the outcome and continuous regressors on a linear time trend before estimation (Table B10).

3.2 Oaxaca–Blinder decomposition

To examine the structural channels through which the post-2022 situation reshaped the composition of labour demand, we decompose the aggregate change in the digital skill share into within- and between-occupation components using a two-period Oaxaca-Blinder shift-share decomposition (Oaxaca 1973; Blinder 1973). The aggregate digital skill share in period t is $S_t = \sum_j w_{jt} s_{jt}$, where w_{jt} is occupation j's share of total vacancies and s_{jt} is its average digital skill share. Using midpoint weights $\bar{w}_j = (w_j^0 + w_j^1)/2$ and $\bar{s}_j = (s_j^0 + s_j^1)/2$, the exact two-period decomposition is:

$$\Delta S = \underbrace{\sum_j \bar{w}_j \, \Delta s_j}_{\text{Within-occupation}} + \underbrace{\sum_j \bar{s}_j \, \Delta w_j}_{\text{Between-occupation}} \tag{6}$$

where superscripts 0 and 1 denote pre- and post-2022 period averages respectively. The within-occupation component captures existing job profiles becoming more digital-skill-intensive at a fixed vacancy mix; the between-occupation component captures demand shifting toward inherently more-digital occupations. Pre-2022 and post-2022 periods are defined as January 2021-January 2022 and February 2022 onwards respectively, with occupation groups following the ISCO-08 one-digit classification.

4 Labor Demand for Digital Skills

Table 2 reports the baseline regression results for the fatalities specification. Column 1 shows the results without controls while Columns 2-4 subsequently add fixed effects. In Columns 5, we add additional controls (vacancy count, ChatGPT rollout, and a dummy for a second AI wave), while in Column 6 we report the fully controlled specification, including the weekly skill share related to military skills. Column 7 further adds the proxy variables for energy disruptions to the set of control variables.

The coefficient on log(1 + fatalities) is positive and statistically significant across all seven specifications, ranging from 0.012^{***} in the baseline to 0.006^{**} under full controls, indicating a robust positive association between attack-related fatalities and the share of job postings requiring digital skills. Column (5) confirms that the coefficient remains strongly significant at 0.010^{***} when controlling for the total number of job postings and the two AI waves, confirming that the positive association is not driven by AI adoption trends or aggregate labour market conditions. The ChatGPT rollout indicator itself enters negatively and significantly (-0.068^{***} in column 3), consistent with AI-induced compression of baseline digital-skill requirements in postings, with the second AI wave indicator reinforcing this pattern (-0.055^{**}). Column (6) shows that the military skill share, which enters positively and significantly (2.149^{***}) and attenuates the coefficient to 0.006^{**}; this pattern is consistent with intensified attacks partly shifting hiring toward defense-adjacent occupations - which might partly require innovative skills, including digital skills. Lastly, Column (7) demonstrates that our results are not confounded by infrastructure disruptions. Table 3 replicates the analysis using attack-related events as the attack measure and yields a virtually identical pattern, confirming robustness to the choice of attack proxy.

To shed light on the economic significance of these associations, we scale the estimated coefficients by the standard deviation of the attack measure. The pre-2022 mean digital skill share was 6.3 percent, rising to 13.5 percent post-2022 - an absolute increase of 7.2 percentage points. In the baseline specification (column 2), a one-standard-deviation increase in log(1 + weekly fatalities) over the full sample - corresponding to a standard deviation of 2.61 log-units, which spans the contrast between the near-zero pre-2022 period and the post-2022 period - is associated with a 3.2 percentage point increase in the digital skill share, equivalent to 51.6 percent of the pre-2022 mean. Restricting variation to the post-2022 period, where week-to-week fluctuations in fatalities are more modest (SD = 0.46 log-units), the implied effect is 0.56 percentage points per standard deviation, or 9.0 percent of the pre-2022 mean. Under full controls (column 6), these estimates attenuate to 1.7 percentage points (27.0 percent of the pre-2022 mean) for full-sample variation and 0.29 percentage points (4.7 percent) for post-2022 variation, providing a conservative lower bound.

The results using log(1 + events) are closely comparable: a full-sample one-standard-deviation increase implies a 3.1 percentage point rise in the baseline specification (49.9 percent of the pre-2022 mean), attenuating to 1.5 percentage points under full controls.

Taken together, these estimates indicate that the association between attack intensity and digital skill demand is economically meaningful relative to the pre-2022 baseline, with the full-sample SD-scaled effects accounting for between one quarter and one half of the pre-2022 mean depending on the specification.

A natural question is whether the effect materializes contemporaneously or with a lag. The moving-average results in Table 4 and Figure 3 show a coefficient of approximately 0.013 percentage points per log unit of fatalities across all three MA windows - 4-week, 8-week, and 12-week - and all four lags, uniformly positive and significant at the 1% level throughout. Under full controls the coefficient attenuates modestly to approximately 0.009-0.011 but remains highly significant. The near-identical estimates across lags and windows reflect the smoothness of the MA series and confirm that the finding is not an artefact of idiosyncratic weekly spikes but reflects a stable medium-run relationship.

Table 2: OLS Regression Results - Log(Fatalities)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log(Fatalities)	0.012***	0.012***	0.013***	0.012***	0.010***	0.006**	0.009**
	(0.001)	(0.001)	(0.003)	(0.001)	(0.003)	(0.003)	(0.004)
Vacancy Count					0.000**	0.000**	0.000**
					(0.000)	(0.000)	(0.000)
Military Skill Share						2.149***	2.111***
						(0.656)	(0.649)
ChatGPT Rollout					−0.017	−0.012	−0.034
					(0.024)	(0.021)	(0.028)
AI Wave 2					0.001	−0.024	−0.021
					(0.024)	(0.022)	(0.026)
Outage Hours (std.)							0.004
							(0.005)
Combat Disconnections (std.)							−0.008
							(0.007)
Intercept	0.049***	0.060***	0.061***	0.061***	0.092***	0.083***	0.082***
	(0.004)	(0.009)	(0.014)	(0.013)	(0.018)	(0.022)	(0.022)
Quarter FE	No	Yes	Yes	No	Yes	Yes	Yes
Year FE	No	No	Yes	No	No	No	No
Month FE	No	No	No	Yes	No	No	No
Partial controls	No	No	No	No	Yes	Yes	Yes
Military share	No	No	No	No	No	Yes	Yes
Energy controls	No	No	No	No	No	No	Yes
Num.Obs.	235	235	235	235	235	235	235
R2	0.302	0.328	0.493	0.363	0.356	0.426	0.436
R2 Adj.	0.299	0.316	0.473	0.328	0.336	0.406	0.411

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: average digital skill share (weekly). Log(Fatalities) is $\log(1 + \text{weekly ACLED fatalities})$. (1) no fixed effects, HC3 standard errors; (2) quarter fixed effects, HAC SE (Newey-West, 4 lags); (3) quarter and year fixed effects, HAC SE; (4) month-of-year fixed effects (12 indicators), HAC SE; (5) partial controls (vacancy count, ChatGPT rollout, AI wave 2) with quarter fixed effects, HAC SE — military skill share excluded; (6) full controls adding military skill share to column (5), HAC SE; (7) column (6) plus energy infrastructure controls: standardised weekly electricity outage hours and standardised combat-related consumer disconnections (MinEnergo), HAC SE. Model (1) uses HC3 SE. Models (2)–(7) use HAC SE with 4 lags.

Table 3: OLS Regression Results - Log(Events)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log(Events)	0.035***	0.036***	0.043***	0.037***	0.027***	0.018*	0.024**
	(0.003)	(0.004)	(0.008)	(0.004)	(0.009)	(0.009)	(0.011)
Vacancy Count					0.000*	0.000**	0.000**
					(0.000)	(0.000)	(0.000)
Military Skill Share						2.231***	2.185***
						(0.674)	(0.664)
ChatGPT Rollout					−0.022	−0.015	−0.037
					(0.026)	(0.022)	(0.032)
AI Wave 2					0.005	−0.022	−0.016
					(0.026)	(0.023)	(0.027)
Outage Hours (std.)							0.006
							(0.005)
Combat Disconnections (std.)							−0.006
							(0.007)
Intercept	−0.113***	−0.105***	−0.132***	−0.109***	−0.027	0.005	−0.024
	(0.016)	(0.026)	(0.038)	(0.028)	(0.061)	(0.062)	(0.065)
Quarter FE	No	Yes	Yes	No	Yes	Yes	Yes
Year FE	No	No	Yes	No	No	No	No
Month FE	No	No	No	Yes	No	No	No
Partial controls	No	No	No	No	Yes	Yes	Yes
Military share	No	No	No	No	No	Yes	Yes
Energy controls	No	No	No	No	No	No	Yes
Num.Obs.	235	235	235	235	235	235	235
R2	0.277	0.307	0.504	0.351	0.344	0.421	0.431
R2 Adj.	0.274	0.295	0.484	0.316	0.323	0.400	0.406

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: average digital skill share (weekly). Log(Events) is log(1 + weekly ACLED events). (1) no fixed effects, HC3 standard errors; (2) quarter fixed effects, HAC SE (Newey-West, 4 lags); (3) quarter and year fixed effects, HAC SE; (4) month-of-year fixed effects (12 indicators), HAC SE; (5) partial controls (vacancy count, ChatGPT rollout, AI wave 2) with quarter fixed effects, HAC SE — military skill share excluded; (6) full controls adding military skill share to column (5), HAC SE; (7) column (6) plus energy infrastructure controls: standardised weekly electricity outage hours and standardised combat-related consumer disconnections (MinEnergo), HAC SE. Model (1) uses HC3 SE. Models (2)–(7) use HAC SE with 4 lags.

Table 4: Moving-Average Smoothing: Single-Lag Specifications

	4-Week MA		8-Week MA		12-Week MA	
	(1)	(2)	(3)	(4)	(5)	(6)
MA log(Fatalities) [lag 0]	0.013***	0.009***	0.013***	0.010***	0.013***	0.011***
	(0.001)	(0.003)	(0.001)	(0.003)	(0.002)	(0.003)
MA log(Fatalities) [lag 1]	0.013***	0.009***	0.013***	0.010***	0.013***	0.011***
	(0.001)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)
MA log(Fatalities) [lag 2]	0.013***	0.009***	0.013***	0.010***	0.013***	0.011***
	(0.001)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)
MA log(Fatalities) [lag 3]	0.013***	0.009***	0.013***	0.010***	0.013***	0.011***
	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)
Observations	232	232	228	228	224	224
R^2	0.355	0.441	0.343	0.439	0.344	0.449
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Full controls	No	Yes	No	Yes	No	Yes

Dependent variable: `avg_digital_skill_share` (weekly). Each row reports a separate OLS model with a single lag of the w-week backward MA of $\log(1 + \text{fatalities})$ as the sole attack-intensity regressor. Odd columns include quarter fixed effects only (baseline). Even columns add vacancy count, military skill share, ChatGPT rollout, and second AI-wave indicators (no linear time trend). HAC standard errors (Newey-West, 4 lags) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The Almon polynomial distributed lag in Table 5 and Figure 4 adds directional precision to the timing question. There is no statistically significant contemporaneous response ($\hat{\beta}_0 = -0.003$ pp, $p > 0.10$). The effect builds from week 1 onward, with positive and marginally significant coefficients at lags 1-3 ($\hat{\beta}_1 = 0.003^{*}$, $\hat{\beta}_2 = 0.004^{*}$, $\hat{\beta}_3 = 0.003^{*}$), before tapering toward zero at lags 4-7. The cumulative effect over the full 0-7 week window is $+0.010^{***}$. This lag profile - no immediate reaction, a positive effect building over weeks 1-3, then tapering - is consistent with firms adjusting job postings with a short operational delay following periods of heightened attack-related activity rather than reacting instantaneously. Together, the moving-average and Almon specifications deliver a coherent picture: the effect operates with a one-to-three-week delay and accumulates meaningfully over a two-month window.

Formal tests suggest the change from February 24 2022 onward is best characterised as

Table 5: Almon Polynomial Distributed Lag: Digital Skill Demand and Fatalities

	PDL (degree 3, lags 0–7)	
	Coef.	(SE)
Log(Fatalities), lag 0 (implied)	-0.003	(0.004)
Log(Fatalities), lag 1 (implied)	0.003*	(0.002)
Log(Fatalities), lag 2 (implied)	0.004*	(0.002)
Log(Fatalities), lag 3 (implied)	0.003*	(0.002)
Log(Fatalities), lag 4 (implied)	0.000	(0.002)
Log(Fatalities), lag 5 (implied)	-0.002	(0.003)
Log(Fatalities), lag 6 (implied)	-0.001	(0.002)
Log(Fatalities), lag 7 (implied)	0.005	(0.003)
Cumulative effect (lags 0–7)	0.010***	(0.003)
Polynomial degree	3	
Observations	228	
R^2	0.444	
Quarter FE	Yes	
Full controls (no trend)	Yes	

Dependent variable: `avg_digital_skill_share` (weekly). Attack intensity measured as log(1 + fatalities). The Almon (1965) polynomial distributed lag constrains lag coefficients β_k of $\log(1+\text{fatalities})$ to lie on a degree-3 polynomial: $\beta_k = \sum_{j=0}^{3} \alpha_j k^j$, $k = 0, 1, \ldots, 7$, spanning approximately two months. The model is estimated by regressing the outcome on the Almon basis variables $Z_{j,t} = \sum_{k=0}^{7} k^j x_{t-k}$ for $j = 0, 1, 2, 3$, plus the same controls as the baseline. Implied lag coefficients and their delta-method standard errors (derived from the HAC variance matrix of the α_j estimates; Newey-West, 4 lags) are reported. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

a discrete regime shift rather than a continuously varying shock. The Chow test ($F = 7.38$, $p < 0.001$), CUSUM test ($p = 0.005$), and Andrews Sup-F test (Sup-F = 94.69, $p < 0.001$) all reject parameter stability, with February 24, 2022 confirmed as a significant break date (see Appendix Table B5 and Figures B6 and B7).

Figure 5 illustrates the pattern: the digital skill share was on a modest declining path before the 24th of February 2022 and jumps discretely upward thereafter. Figure 6 overlays fitted values from an ITS model - which allows both intercept and slope to shift at the targeted date - against the continuous-regressor baseline, showing that the ITS model tracks

Figure 3: Moving-average smoothing specification

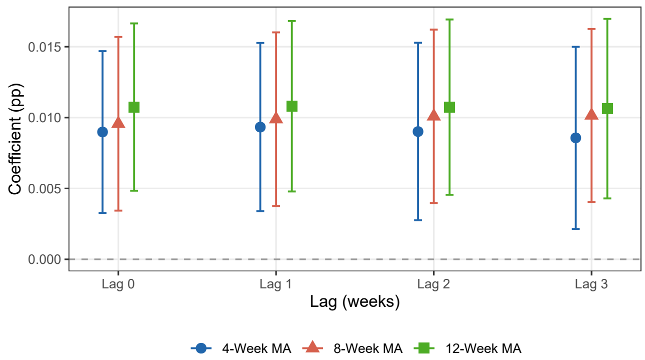
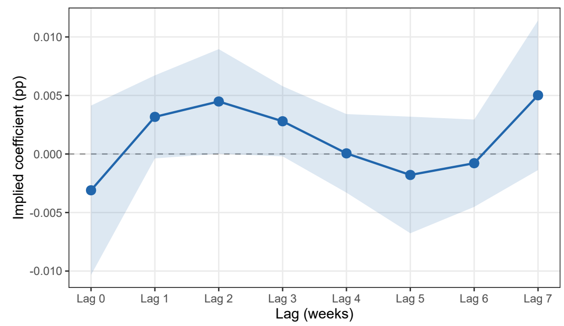


Figure 4: Almon polynomial distributed lag specification



Coefficient plots: moving-average smoothing and Almon distributed lag specifications. Left panel: coefficient on the w-week backward MA of $\log(1 + \text{fatalities})$ at lags 0-3, for 4-week, 8-week, and 12-week windows. Right panel: implied lag coefficients from the Almon PDL model (degree 3, lags 0-7; cumulative effect $+0.010^{***}$, see Table 5). All estimates use full controls. Bars and shaded bands denote 95% confidence intervals based on HAC standard errors (Newey-West, 4 lags).

the observed discontinuity closely while the continuous baseline approximates it as a smooth gradient.

The ITS specification in Table 6 yields a post-2022 coefficient that is large, positive, and highly significant across all control configurations ($+0.052^{***}$ to $+0.069^{***}$). The drop in the coefficient in column (2), which adds military skill share and vacancy count, reflects the fact that both variables are themselves partly attack-induced, making that column a conservative lower bound. Critically, when both the ITS indicators and log(fatalities) are included simultaneously (Appendix Table B6, Model 5), the continuous regressor becomes negligible (-0.004, $p > 0.05$) while the post-2022 dummy remains large and significant, indicating that the continuous intensity measure was capturing the structural break rather than independent week-to-week variation in attack intensity. The explanatory power of the structural break specification ($R^2 = 0.464$) matches the continuous baseline, confirming the two approaches are substantively equivalent. We interpret this as evidence of a post-2022-driven regime shift in employer skill demand.

Table 6: ITS Robustness: Digital Skill Share with Progressive Controls

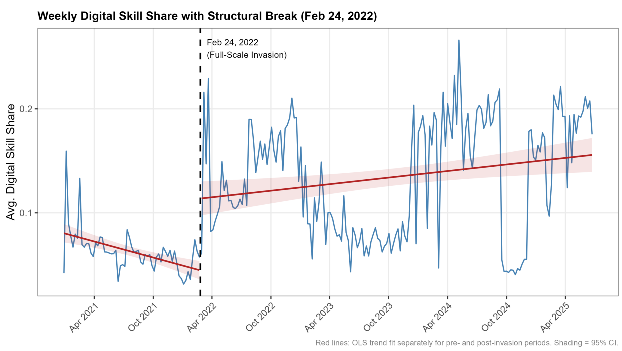
	(1) ITS Baseline	(2) ITS + Econ. Controls	(3) ITS + Extended Controls	(4) ITS + AI Wave 2	(5) ITS + Energy
Post-2022	0.052***	0.028*	0.056***	0.048***	0.069***
	(0.014)	(0.016)	(0.019)	(0.018)	(0.019)
Time Since February 2022	0.000*	0.000	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Vacancy Count		0.000*	0.000*	0.000*	0.000*
		(0.000)	(0.000)	(0.000)	(0.000)
Military Skill Share		1.477**	0.515	0.417	0.515
		(0.733)	(0.711)	(0.658)	(0.626)
Time Trend			0.000	0.000*	0.000
			(0.000)	(0.000)	(0.000)
ChatGPT Rollout			−0.068***	−0.039	−0.062**
			(0.021)	(0.026)	(0.031)
AI Wave 2 (Claude/GPT-4/Bard)				−0.055**	−0.051*
				(0.025)	(0.028)
Outage Hours (std.)					0.002
					(0.005)
Combat Disconnections (std.)					−0.009
					(0.006)
Intercept	0.073***	0.087***	0.112***	0.110***	0.105***
	(0.008)	(0.018)	(0.017)	(0.018)	(0.017)
Num.Obs.	235	235	235	235	235
R2	0.353	0.387	0.464	0.496	0.504
R2 Adj.	0.339	0.368	0.442	0.474	0.477

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

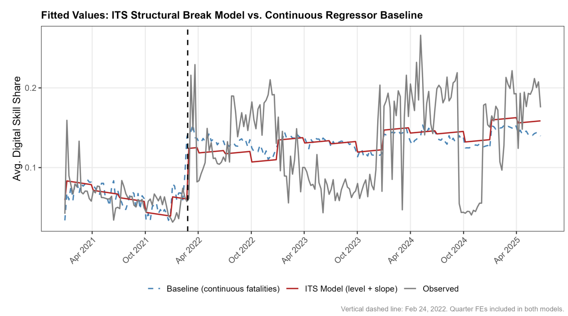
Dependent variable: average digital skill share. All models use an Interrupted Time Series (ITS) design with a discrete post-2022 indicator in place of the continuous log(fatalities) regressor. The discrete shock on February 24, 2022 is exogenous to gradual, time-varying confounders. (1) ITS baseline + quarter FE; (2) adds vacancy count and military skill share; (3) extended controls: adds time trend and ChatGPT rollout; (4) additionally controls for the second AI wave (Claude 1.0, GPT-4, and Google Bard, all launched in the week of March 14, 2023); (5) additionally controls for energy infrastructure disruptions: standardised weekly electricity outage hours (energy-map.info, from January 2023; pre-2023 set to zero) and standardised combat-related consumer disconnections (Ukrainian Ministry of Energy, from March 2022; pre-2022 set to zero). HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included but not shown.

Figure 5: Weekly digital skill share with structural break (Feb 24, 2022).



Red lines: OLS trend fit separately for pre- and post-2022 periods. Shading = 95% CI. Vertical dashed line: 24 February 2022. Based on Jooble data (Jan, 2021-June, 2025).

Figure 6: Fitted values from the Interrupted Time Series (ITS) structural break model versus the continuous regressor baseline and observed values.



The ITS model (red solid line) allows both an intercept and slope shift at the targeted date (24 February 2022); the baseline (blue dashed line) uses log(1 + fatalities) as a continuous regressor. Both models include quarter fixed effects. The ITS model tracks the discrete jump at the targeted date closely, while the continuous baseline approximates the break as a smooth gradient. Vertical dashed line: 24 February 2022. Based on Jooble data (Jan, 2021-June, 2025).

Our results are consistent with Hershbein and Kahn (2018), who document that economic disruptions accelerate skill-biased technological change, and with Pham, Talavera, and Wu (2023), who find task composition shifts in Ukrainian vacancies due to the current situation, though their skill-level approach differs from our digital skill measure. Moreover, the discrete regime shift at the targeted date is consistent with Forsythe et al. (2020), who document an abrupt compositional shift in vacancy skill requirements at the onset of the COVID-19 pandemic, suggesting that large discrete shocks - whether health crises or military interventions - generate structural breaks in labor demand rather than smooth adjustments.

5 Robustness

5.1 Frequency Level

A potential concern with the weekly analysis is that high-frequency variation in both job postings and attack events may reflect short-run noise rather than deliberate firm-level adjustments to hiring strategy. To address this, we re-estimate the main OLS specifications using monthly-aggregated data. Attack intensity is measured as $\log(1{+}\text{monthly sum of ACLED fatalities})$ and $\log(1 + \text{monthly event count})$, respectively. The sample spans 54 calendar months from January 2021 through June 2025.

The monthly results are reported in Tables B3 and B4. Across all four specifications - (1) no fixed effects with HC3 standard errors, (2) quarter fixed effects, (3) quarter-and-year fixed effects, and (4) month-of-year fixed effects with Newey-West standard errors (3 lags) - the coefficient on attack intensity remains positive and statistically significant at the 1 percent level. In the fatalities specification, the point estimate is stable at approximately 0.011 across specifications (1), (2), and (4), rising modestly to 0.014 when year fixed effects are added. The events specification yields similarly stable estimates of 0.033-0.034, with the same modest amplification under quarter-and-year fixed effects (0.050). The sign, magnitude, and significance of the monthly estimates closely mirror their weekly counterparts, providing no evidence that the main findings are driven by high-frequency noise.

5.2 Addressing Spurious Correlation

Unit root tests confirm that both the digital skill share and log fatalities are integrated of order one, I(1) (Appendix Table B7). For the digital skill share, the ADF test fails to reject the unit root null ($p = 0.070$), the KPSS Level test strongly rejects stationarity ($p < 0.01$), and the KPSS Trend test rejects at the 10% level ($p = 0.086$); for log fatalities, all tests consistently indicate a unit root. This raises the possibility of spurious regression.

We address this by testing for cointegration and estimating Error Correction Models (ECMs). Appendix B presents the methodological details. The Engle-Granger two-step procedure (Appendix Table B8) yields stationary residuals from the full long-run specification (ADF = -3.934, $p = 0.013$), indicating a cointegrating relationship. The ECM error-correction term is negative and highly significant across all specifications ($\hat{\gamma} \approx -0.33$ to -0.43, $p < 0.001$; Appendix Table B9), implying that approximately one-third of any deviation from the long-run equilibrium is eliminated within a single week. The insignificance of the short-run Δ coefficients is consistent with skill demand responding to structural changes in attack intensity over time rather than to week-to-week fluctuations.

As a complementary check under the trend-stationary alternative, Table B10 residualises both the outcome and continuous regressors on a linear time trend before estimation; the detrended log(fatalities) coefficient remains positive and significant (0.008^{***}), and the post-2022 ITS indicator likewise (0.038^{***}), confirming the relationship is not an artefact of common deterministic trends. Once the full control set - vacancy count, military skill share, and the ChatGPT rollout - is added alongside detrending (column 3), both the continuous regressor and the post-2022 indicator attenuate toward zero and lose significance. We interpret this not as evidence against the main finding, but as a consequence of combining two trend-removal strategies simultaneously: detrending removes the deterministic component shared by the outcome and regressors, while military skill share and vacancy count - shown in Section 4 to be themselves partly attack-induced and collinear with the post-2022 trend - absorb much of what remains. Stacking these together leaves little independent variation for log(fatalities) to explain, and the sharp drop in R^2 relative to the corresponding non-detrended specification (from around 0.4 in Table 2, column 6, to 0.259 here) is consistent with a loss of power rather than a genuine null result. We therefore view columns (1) and (2) as the cleaner tests of the trend-stationary alternative, and note that the full-controls specification is, by construction, the most conservative test in the paper.

5.3 Decomposition of the Digital Skill Share

To assess whether the observed rise in the digital skill share reflects genuine upskilling demand or a mechanical compositional effect driven by labour market contraction, we regress the raw count of digital skill postings - $\log(1 + \text{total digital skills})$ - on the same explanatory variables used in the main analysis (Appendix Table B11). The ITS specification (Column 2) yields a large, precisely estimated negative effect: the post-2022 indicator is associated with a decline of 2.557 log-points in absolute digital posting volumes ($p < 0.001$), confirming that the post-2022 situation suppressed digital labour demand in absolute terms. The Time Since February 2022 coefficient in the full-controls specification (Column 3) is positive and significant (0.010, $p < 0.05$), suggesting a gradual recovery in absolute digital hiring in subsequent weeks even as the share continued to rise.

The rise in the digital skill share therefore reflects the relative resilience of digital hiring: total vacancy volumes contracted more sharply than digital postings. Table B12 shows that the post-2022 indicator is associated with a large, statistically significant decline in total vacancy postings (2.651 log-points, $p < 0.01$), while the corresponding estimate for digital postings is smaller in magnitude and statistically insignificant (0.615, $p > 0.1$). Figure B8 illustrates the pattern visually. We interpret our main findings accordingly - as evidence of an attack-induced compositional shift in the skill structure of labour.

6 Mechanisms

This section investigates the channels through which the post-2022 situation raised digital skill demand. We examine two complementary mechanisms: the adoption of remote work arrangements, which are inherently contingent on digital proficiency, and shifts in occupational composition toward more digital-intensive roles.

Tables 7 and 8 explore the remote work channel by progressively adding remote share to the baseline specification for fatalities and events respectively. Column (1) reproduces the baseline results (0.012^{***} and 0.036^{***}). Column (2) adds vacancy count, ChatGPT rollout, and AI wave 2 indicators, leaving the attack coefficients largely unchanged (0.010^{***} and 0.027^{***}). Column (3) adds the weekly remote work share, which enters strongly and precisely estimated in both specifications (0.354^{***} and 0.351^{***}), with R^2 rising sharply from approximately 0.35 to 0.895-0.896. The respective coefficients attenuate to zero once remote share is controlled for, consistent with remote work adoption being a key structural channel linking the post-2022 situation to higher digital skill demand. That the first-stage regression (Table B13) of remote share on log(fatalities) confirms a strong positive relationship ($\hat{\beta} = 0.041^{***}$, $p < 0.001$, $R^2 = 0.432$) further supports this interpretation: remote work itself rose sharply and persistently as a direct consequence of the current situation in Ukraine.

Remote share should nonetheless be understood as a *mediating mechanism* rather than a confounder. It is a consequence of the post-2022 situation, not a pre-existing trend that would have shifted digital skill demand independently of the break experienced since February 2022. Including it as a control therefore introduces bad-control bias (Angrist and Pischke 2009) - the remote share column provides a lower bound on the attack effect rather than a preferred estimate. Column (4) further adds military skill share, which is statistically insignificant in both specifications once remote share is controlled for (0.101 and 0.054, both $p > 0.10$), suggesting that the remote work channel is the dominant pathway through which the post-2022 situation raised digital skill demand. The strong mediating role of remote work adoption is consistent with Forsythe et al. (2020), who document a sharp rise in remote work postings

Table 7: Remote Work as a Mediating Channel: Digital Skill Demand and Fatalities

	(1) Baseline	(2) + Controls	(3) + Remote	(4) + Military
Log(Fatalities)	0.012***	0.010***	0.000	0.000
	(0.001)	(0.003)	(0.001)	(0.001)
Remote Share			0.354***	0.352***
			(0.013)	(0.014)
Vacancy Count		0.000**	0.000**	0.000**
		(0.000)	(0.000)	(0.000)
Military Skill Share				0.101
				(0.232)
ChatGPT Rollout		−0.017	−0.003	−0.002
		(0.024)	(0.009)	(0.008)
AI Wave 2		0.001	−0.027***	−0.028***
		(0.024)	(0.009)	(0.009)
Intercept	0.060***	0.092***	0.046***	0.046***
	(0.009)	(0.018)	(0.007)	(0.008)
Num.Obs.	235	235	235	235
R2	0.328	0.356	0.895	0.895
R2 Adj.	0.316	0.336	0.891	0.891

* p < 0.1, ** p < 0.05, *** p < 0.01

Dependent variable: average digital skill share (weekly). All models include quarter fixed effects and HAC standard errors (Newey-West, 4 lags). (1) quarter FE only; (2) adds vacancy count, ChatGPT rollout, and AI wave 2; (3) adds remote share to column (2); (4) further adds military skill share. Remote share is the weekly share of postings explicitly mentioning remote or distance work. Since remote adoption is itself partly attack-induced, column (3) provides a lower bound on the fatalities effect (bad-control bound).

at the onset of COVID-19, and with the broader evidence that remote work is contingent on digital proficiency (Acemoglu et al. 2021).

We next examine compositional effects through an Oaxaca-Blinder decomposition of the change in the vacancy-weighted average digital skill share into *within-occupation* and *between-occupation* components. The aggregate share is $S_t = \sum_j w_{jt} s_{jt}$, where w_{jt} is occupation j's share of total vacancies in month t and s_{jt} is its average digital skill share. Using midpoint weights, the exact two-period decomposition is:

$$\Delta S = \underbrace{\sum_j \bar{w}_j \, \Delta s_j}_{\text{Within-occupation}} + \underbrace{\sum_j \bar{s}_j \, \Delta w_j}_{\text{Between-occupation}} \tag{7}$$

Table 8: Remote Work as a Mediating Channel: Digital Skill Demand and Events

	(1) Baseline	(2) + Controls	(3) + Remote	(4) + Military
Log(Events)	0.036***	0.027***	0.004	0.004
	(0.004)	(0.009)	(0.004)	(0.004)
Remote Share			0.351***	0.350***
			(0.013)	(0.014)
Vacancy Count		0.000*	0.000	0.000
		(0.000)	(0.000)	(0.000)
Military Skill Share				0.054
				(0.233)
ChatGPT Rollout		−0.022	−0.005	−0.005
		(0.026)	(0.009)	(0.009)
AI Wave 2		0.005	−0.027***	−0.027***
		(0.026)	(0.009)	(0.009)
Intercept	−0.105***	−0.027	0.019	0.020
	(0.026)	(0.061)	(0.023)	(0.023)
Num.Obs.	235	235	235	235
R2	0.307	0.344	0.896	0.896
R2 Adj.	0.295	0.323	0.892	0.892

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: average digital skill share (weekly). Log(Events) is log(1 + weekly ACLED events). All models include quarter fixed effects and HAC standard errors (Newey-West, 4 lags). (1) quarter FE only; (2) adds vacancy count, ChatGPT rollout, and AI wave 2; (3) adds remote share to column (2); (4) further adds military skill share. Remote share is the weekly share of postings explicitly mentioning remote or distance work. Since remote adoption is itself partly attack-induced, column (3) provides a lower bound on the post-2022 effect (bad-control bound).

where $\bar{w}_j = (w_j^0 + w_j^1)/2$, $\bar{s}_j = (s_j^0 + s_j^1)/2$, and superscripts 0 and 1 denote pre- and post-2022 period averages. The within component captures occupations becoming more digital-skill-intensive at a fixed vacancy mix; the between component captures demand shifting toward inherently more-digital occupations. Figure 7 illustrates that the composition-only counterfactual - holding occupation-specific digital intensity fixed at 2021 levels - remains substantially below the observed series throughout the post-2022 period, indicating that both channels are active.

Figure 7: Digital skills and occupational changes in Ukraine (2021-2025)

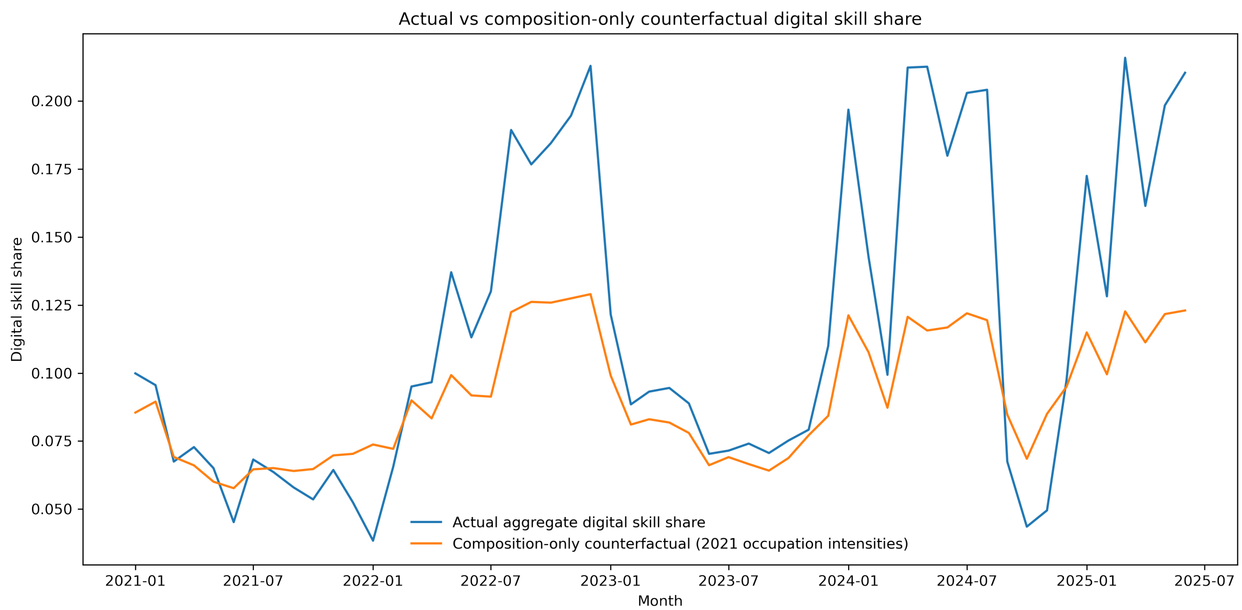


Table 9 reports the results. The aggregate digital skill share rose by 6.35 percentage points from the pre-2022 period (January 2021-January 2022) to the post-2022 period (February 2022 onward), split approximately equally between a between-occupation component of 3.31 pp (52.1%) and a within-occupation component of 3.05 pp (48.0%). Figure 8 shows that both components are near zero before February 2022 and rise persistently thereafter, consistent with a attack-driven structural shift rather than a pre-existing trend.

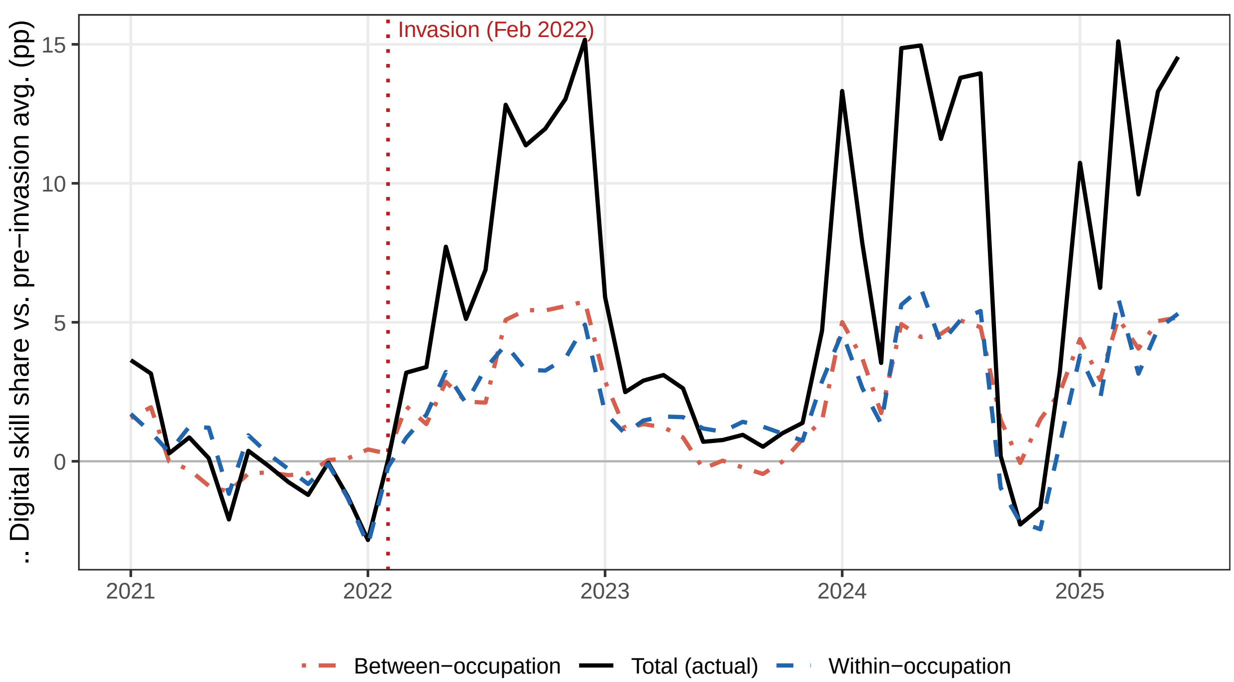
The occupation-level breakdown in Figure 9 reveals that *Professionals* account for the overwhelming share of both components: their vacancy share nearly doubled from 33.9% to

Table 9: Within- and Between-Occupation Decomposition of Digital Skill Share

	Dig. share (%)		Vac. share (%)		Contribution (pp)		
Occupation	Pre	Post	Pre	Post	Within	Between	Total
Professionals	15.79	21.77	33.93	52.19	2.58	3.43	6.00
Technicians and associate professionals	4.08	6.84	13.60	9.27	0.32	-0.24	0.08
Service and sales workers	0.71	0.63	12.14	9.57	-0.01	-0.02	-0.03
Craft and related trades workers	1.89	1.84	13.47	6.76	-0.01	-0.13	-0.13
Plant and machine operators and assemblers	1.32	1.43	8.61	5.14	0.01	-0.05	-0.04
Managers	5.05	7.00	3.14	9.75	0.13	0.40	0.52
Elementary occupations	0.71	0.69	9.10	3.30	-0.00	-0.04	-0.04
Clerical support workers	2.59	3.35	5.66	3.91	0.04	-0.05	-0.02
Skilled agricultural, forestry and fishery workers	0.20	0.69	0.35	0.18	0.00	-0.00	0.00
Total	**6.74**	**13.09**	**100.00**	**100.00**	**3.05**	**3.31**	**6.35**

Midpoint (Oaxaca-Blinder) shift-share decomposition. *Within* = change in aggregate digital skill share holding occupation vacancy shares fixed at their midpoint values; it captures occupations becoming more digital-skill-intensive. *Between* = change holding occupational digital content fixed; it captures demand shifting toward inherently more-digital occupations. Within + Between = Total (exact with midpoint weights). Occupation groups follow ISCO-08 1-digit classification. Pre-2022 reference: Jan 2021–Jan 2022; post-2022: Feb 2022–latest available month. Figures in % are averages within the respective period.

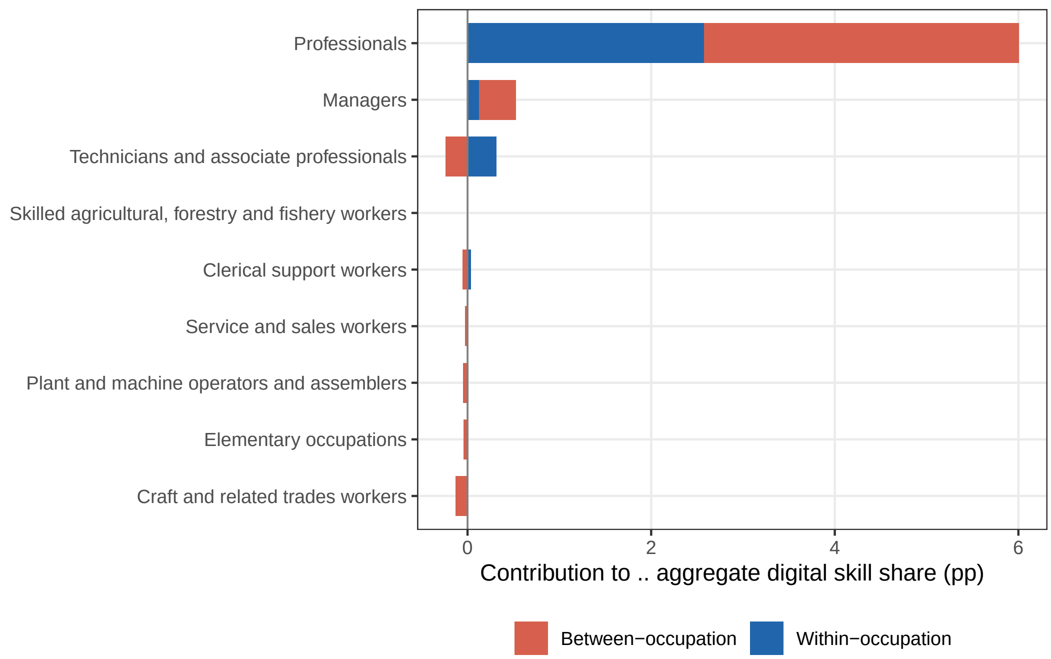
Figure 8: Occupation-level contributions to the aggregate change in digital skill share (pp), decomposed into within- and between-occupation components



Notes: Vertical dotted line marks February 2022.

52.2%, while their average digital skill share also rose (15.8% → 21.8%), yielding a combined contribution of +6.00 pp - 94.5% of the total aggregate change. *Managers* contribute a further +0.52 pp, driven primarily by a rising vacancy share. All remaining occupation groups exhibit small negative between-occupation contributions, reflecting contraction in their share of the vacancy pool, though within-occupation upgrades partially offset these losses for Technicians and Plant operators.

Figure 9: Occupation-level contributions to the aggregate change in digital skill share (pp), decomposed into within- and between-occupation components



Notes: Two-period Oaxaca-Blinder decomposition; ISCO-08 1-digit classification. Total aggregate change: +6.35 pp.

The dominance of the between-occupation component - demand shifting toward Professionals and Managers - echoes Autor, Levy, and Murnane (2003) and Acemoglu and Restrepo (2018), who predict that technological disruptions reallocate labor toward non-routine cognitive occupations. That roughly half the shift is within-occupation is consistent with Hershbein and Kahn (2018), who find that recessions induce upskilling within existing job categories rather than purely reallocating workers across occupations.

Taken together, the two mechanisms are complementary. Remote work adoption - itself a direct response to attack-induced barriers to on-site work - emerges as the dominant channel linking attack intensity to digital skill demand on a week-to-week basis. The occupational decomposition reveals the structural counterpart: the post-2022 situation simultaneously tilted the composition of labour demand toward high-skill, digital-intensive occupations, most prominently Professionals, while also inducing within-occupation upskilling across existing job profiles. These two channels, operating at different frequencies and through different firm-level decisions, together account for the related regime shift in digital skill demand documented in Section 5.

7 Conclusion

This paper examines how invasion-related disruption shapes the demand for digital skills, contributing to a limited but growing literature on war, conflict, and invasion and labor market adjustment. Using a dataset of more than 17 million online vacancies in Ukraine - classified with large language models and mapped to the ESCO taxonomy, then merged with ACLED measures of attack-related intensity - we document a robust positive association between weekly fatalities and the share of postings requiring digital skills.

The results hold when accounting for multiple forms of seasonality, applying moving-average smoothing, estimating Almon polynomial lag specifications, and controlling for broader labor-market conditions, including aggregate vacancy volume and the roll-out of ChatGPT and a second wave of LLMs. We further show that part of the increase in digital-skill demand is linked to shifts in occupational composition, an increase in digital skill demand within existing occupations, and a shift to remote work.

Taken together, the results suggest that firms may respond to attack-induced labor scarcity, asset damage, and uncertainty by shifting toward digitally intensive production and coordination, leading to increased labor demand for digital skills. We caution, however, that our estimates are associative rather than causal, and rely on online vacancy data that may under-represent the public and informal sectors and that LLM classification, while scalable, is not error-free.

The within-occupation upskilling component of the decomposition - whereby existing job profiles became more digitally intensive, accounting for roughly half of the total increase in digital skill share - and the remote work adoption evidence together suggest that genuine behavioral adaptation is occurring alongside compositional shifts. This has two concrete implications. First, reskilling programs targeted at workers displaced from physical or routine-task-intensive roles are likely to find real employer demand - the demand signal is broad-based across existing job categories rather than concentrated in specialist IT roles, suggesting that foundational digital skills training is more relevant than advanced technical upskilling for most displaced workers. Second, remote work adoption emerges as the dominant mediating channel linking attack intensity to digital skill demand, implying that investments in reliable connectivity, electricity continuity, and cloud-based work platforms are not purely welfare measures but productivity interventions with measurable labor market returns that allow firms to maintain hiring even under active combat conditions.

At the same time, the between-occupation component of the decomposition - and the absolute decline in non-digital vacancy postings - suggest that the destruction of non-digital sectors is a significant driver of the observed pattern. The sharp contraction in craft trades, plant operations, and elementary occupations signals an emerging skills gap in precisely the physical labor categories that post-2022 reconstruction will require most urgently. Policymakers should therefore ensure that workforce development strategies are not narrowly focused on digital upskilling at the expense of rebuilding capacity in manufacturing, construction, and other non-digital sectors. A balanced approach - combining digital skills investment with active labor market programs targeting non-digital sector recovery - is likely to yield higher aggregate employment gains in the post-2022 period than a digital-only strategy.

Finally, our findings demonstrate that online vacancy data processed with large language models and mapped to a standard skills taxonomy can provide near-real-time labor market intelligence even in active combat settings, where traditional labor force surveys are difficult to administer and subject to severe coverage gaps. The analytical pipeline developed in this paper offers a model for continuous labor market monitoring - improving the evidence base for reconstruction workforce planning well before reconstruction begins.

While the Ukraine context is specific, the mechanisms we identify - remote work adoption, occupational recomposition, and within-occupation upskilling - are likely to generalize to other related settings, offering actionable lessons for workforce planning during destruction and reconstruction periods. Strengthening digital infrastructure and human capital may help sustain employment and productivity during these types of settings and accelerate recovery once conditions stabilize.

References

Acemoglu, Daron and David Autor (2011). "Skills, tasks and technologies: Implications for employment and earnings". In: *Handbook of Labor Economics*. Vol. 4. Elsevier, pp. 1043–1171.

Acemoglu, Daron and Pascual Restrepo (2018). "The race between man and machine: Implications of technology for growth, factor shares, and employment". In: *American economic review* 108.6, pp. 1488–1542.

— (2019). "Automation and new tasks: How technology displaces and reinstates labor". In: *Journal of Economic Perspectives* 33.2, pp. 3–30.

— (2020). "The wrong kind of AI? Artificial intelligence and the future of labor demand". In: *Cambridge Journal of Regions, Economy and Society* 13.1, pp. 25–35.

— (2022). "Demographics and automation". In: *The Review of Economic Studies* 89.1, pp. 1–44.

Acemoglu, Daron et al. (2021). “AI and jobs: Evidence from online vacancies”. In: *Journal of Labor Economics* 39.1, pp. 1–38.

ACLED, Armed Conflict Location & Event Data Project (2023). *ACLED Dataset, Version 24.1*. https://www.acleddata.com. Accessed: 2025-08-20.

Akcigit, Ufuk et al. (2025). *Engineering Ukraine’s Wirtschaftswunder*. Working Paper 34103. National Bureau of Economic Research. DOI: 10.3386/w34103.

Almon, Shirley (1965). “The Distributed Lag Between Capital Appropriations and Expenditures”. In: *Econometrica* 33.1, pp. 178–196.

Andrews, Donald W. K. (1993). “Tests for Parameter Instability and Structural Change with Unknown Change Point”. In: *Econometrica* 61.4, pp. 821–856.

Angrist, Joshua D. and Jörn-Steffen Pischke (2009). *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton, NJ: Princeton University Press.

Anosike, Paschal (2018). “Entrepreneurship education knowledge transfer in a conflict Sub-Saharan African context”. In: *Journal of small business and enterprise development* 25.4, pp. 591–608.

Autor, David H (2015). “Why are there still so many jobs? The history and future of workplace automation”. In: *Journal of economic perspectives* 29.3, pp. 3–30.

Autor, David H, Frank Levy, and Richard J Murnane (2003). “The skill content of recent technological change: An empirical exploration”. In: *The Quarterly journal of economics* 118.4, pp. 1279–1333.

Barrero, Jose Maria, Nicholas Bloom, and Steven J. Davis (2021). “Why Working from Home Will Stick”. In: *National Bureau of Economic Research Working Paper* 28731. DOI: 10.3386/w28731.

Bauer, Michal et al. (2016). “Can War Foster Cooperation?” In: *Journal of Economic Perspectives* 30.3, pp. 249–274. DOI: 10.1257/jep.30.3.249.

Blattman, Christopher and Edward Miguel (2010). “Civil War”. In: *Journal of Economic Literature* 48.1, pp. 3–57. DOI: 10.1257/jel.48.1.3.

Blinder, Alan S. (1973). “Wage Discrimination: Reduced Form and Structural Estimates”. In: *Journal of Human Resources* 8.4, pp. 436–455.

Blumenstock, Joshua E. et al. (2024). “Violence and Financial Decisions: Evidence from Mobile Money in Afghanistan”. In: *Review of Economics and Statistics* 106.2, pp. 352–369. DOI: 10.1162/rest_a_01147.

Box, George E. P. and George C. Tiao (1975). “Intervention Analysis with Applications to Economic and Environmental Problems”. In: *Journal of the American Statistical Association* 70.349, pp. 70–79.

Chauvin, Nicolas Depetris and Dominic Rohner (2009). *The Effects of Conflict on the Structure of the Economy*. ZBW.

Chiovelli, Giorgio, Stelios Michalopoulos, and Elias Papaioannou (2025). “Landmines and Spatial Development”. In: *Econometrica* 93.5, pp. 1739–1778. DOI: 10.3982/ECTA17951.

Clemens, Jeffrey and Parker Rogers (2023). “Demand shocks, procurement policies, and the nature of medical innovation: Evidence from wartime prosthetic device patents”. In: *Review of Economics and Statistics*, pp. 1–45.

Collier, Paul and Marguerite Duponchel (2013). “The economic legacy of civil war: firm-level evidence from Sierra Leone”. In: *Journal of Conflict Resolution* 57.1, pp. 65–88.

Diamond, Peter (2011). “Unemployment, vacancies, wages”. In: *American Economic Review* 101.4, pp. 1045–1072.

Dickey, David A. and Wayne A. Fuller (1979). “Distribution of the Estimators for Autoregressive Time Series with a Unit Root”. In: *Journal of the American Statistical Association* 74.366, pp. 427–431.

Dingel, Jonathan I. and Brent Neiman (2020). “How Many Jobs Can be Done at Home?” In: *Journal of Public Economics* 189, p. 104235. DOI: 10.1016/j.jpubeco.2020.104235.

EAPL (2025). *Ukrainian Online Marriages and their Recognition in Poland*. https://eapil.org/2025/06/03/ukrainian-online-marriages-and-their-recognition-in-poland/. Accessed: 2025-08-26.

EGA (2024). *Expansion of the e-Entrepreneur service empowers business in Ukraine*. https://ega.ee/expansion-e-entrepreneur-ukraine/. Accessed: 2025-08-26.

Eloundou, Tyna et al. (2024). "GPTs are GPTs: Labor market impact potential of LLMs". In: *Science* 384.6702, pp. 1306–1308. DOI: 10.1126/science.adj0998.

Engle, Robert F. and C. W. J. Granger (1987). "Co-integration and Error Correction: Representation, Estimation, and Testing". In: *Econometrica* 55.2, pp. 251–276.

EU4Digital (2024). *New digital services for Ukraine: Diia Summit 2023*. https://eufordigital.eu/new-digital-services-for-ukraine-diia-summit-2023/. Accessed: 2025-08-26.

Forsythe, Eliza et al. (2020). "Labor demand in the time of COVID-19: Evidence from vacancy postings and UI claims". In: *Journal of public economics* 189, p. 104238.

Granger, C. W. J. and Paul Newbold (1974). "Spurious Regressions in Econometrics". In: *Journal of Econometrics* 2.2, pp. 111–120.

Gregory, Terry, Anna Salomons, and Ulrich Zierahn (2022). "Racing With or Against the Machine? Evidence on the Role of Trade in Technology and Structural Change". In: *Journal of the European Economic Association* 20.2, pp. 861–905. DOI: 10.1093/jeea/jvab040.

Gross, Daniel P and Bhaven N Sampat (2023). "America, jump-started: World War II R&D and the takeoff of the US innovation system". In: *American Economic Review* 113.12, pp. 3323–3356.

Hershbein, Brad and Lisa B Kahn (2018). "Do recessions accelerate routine-biased technological change? Evidence from vacancy postings". In: *American Economic Review* 108.7, pp. 1737–1772.

Jung, Yeonha, Gedeon Lim, and Sangyoon Park (2025). *Civilian Killings and Long-Run Development: Evidence from the Korean War*. Working Paper 11974. CESifo.

Kurekova, Lucia M., Miroslav Beblavy, and Anna Thum-Thysen (2015). "Using Online Vacancies and Web Surveys to Analyse the Labour Market: A Methodological Inquiry". In: *IZA Journal of Labor Economics* 4.18. DOI: 10.1186/s40172-015-0033-2.

Kwiatkowski, Denis et al. (1992). "Testing the Null Hypothesis of Stationarity against the Alternative of a Unit Root". In: *Journal of Econometrics* 54.1–3, pp. 159–178.

Kyiv Global Government Technology Center (2025a). *Diia Ecosystem: the digital milestone in Ukraine*. https://www.kyivgovtechcentre.org/diia-ecosystem. Accessed: 2025-08-26.

— (2025b). *Ukrainian Business Digitalization Index*. https://www.kyivgovtechcentre.org/report_on_the_level_of_business_digitalization_in_ukraine_published. Accessed: 2025-08-26.

Maltsev, Vladimir (2023). "The economics of military innovation under anarchy: The case of the Ukrainian Civil War of 1917–1921". In: *Journal of Economic Behavior & Organization* 210, pp. 180–190.

Michaels, Guy, Ashwini Natraj, and John Van Reenen (2014). "Has ICT Polarized Skill Demand? Evidence from Eleven Countries over Twenty-Five Years". In: *Review of Economics and Statistics* 96.1, pp. 60–77. DOI: 10.1162/REST_a_00366.

Modestino, Alicia Sasser, Daniel Shoag, and Joshua Ballance (2016). "Upskilling: Do Employers Demand Greater Skill When Workers Are Plentiful?" In: *Review of Economics and Statistics* 98.4, pp. 793–805. DOI: 10.1162/REST_a_00538.

— (2020). "Downskilling: Changes in Employer Skill Requirements over the Business Cycle". In: *Labour Economics* 66, p. 101887. DOI: 10.1016/j.labeco.2020.101887.

Napierala, Joanna, Vladimir Kvetan, and Jiri Branka (2022). *Assessing the Representativeness of Online Job Advertisements*. Cedefop Working Paper 17. European Centre for the Development of Vocational Training (Cedefop). DOI: 10.2801/79939.

Newey, Whitney K. and Kenneth D. West (1987). "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix". In: *Econometrica* 55.3, pp. 703–708.

Oaxaca, Ronald (1973). "Male-Female Wage Differentials in Urban Labor Markets". In: *International Economic Review* 14.3, pp. 693–709.

Pham, Tho, Oleksandr Talavera, and Zhuangchen Wu (2023). "Labor markets during war time: Evidence from online job advertisements". In: *Journal of comparative economics* 51.4, pp. 1316–1333.

Phillips, Peter C. B. and Pierre Perron (1988). "Testing for a Unit Root in Time Series Regression". In: *Biometrika* 75.2, pp. 335–346.

Prem, Mounu, Miguel E. Purroy, and Juan F. Vargas (2025). "Landmines: The Local Effects of Demining". In: *Journal of Public Economics* 247, p. 105399. DOI: 10.1016/j.jpubeco.2025.105399.

Reimers, Nils and Iryna Gurevych (2019). "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pp. 3982–3992. DOI: 10.18653/v1/D19-1410.

Riaño, Juan Felipe and Felipe Valencia Caicedo (2024). "Collateral Damage: The Legacy of the Secret War in Laos". In: *The Economic Journal* 134.661, pp. 2101–2140. DOI: 10.1093/ej/ueae004.

Safir, Abla and Noel Muller (2019). *What Employers Actually Want: Skills in Demand in Online Job Vacancies in Ukraine*. Social Protection and Jobs Discussion Paper 1932. World Bank. URL: https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099725006062228836.

Sarioglo, Volodymyr and Oleksandr Cymbal (2020). *Labour Market Landscaping: Ukraine. Big Data for Labour Market Intelligence Project*. Tech. rep. European Training Foundation.

Solow, Robert M (1956). "A contribution to the theory of economic growth". In: *The quarterly journal of economics* 70.1, pp. 65–94.

Turrell, Arthur et al. (2019). *Transforming naturally occurring text data into economic statistics: The case of online job vacancy postings*. Tech. rep. National Bureau of Economic Research.

Udomsaph, Charles (2023). “The Paradox of Innovation in Conflict Zones”. In.

Verwimp, Philip, Patricia Justino, and Tilman Brück (2019). *The microeconomics of violent conflict*.

World Bank (2023). *Ukraine: Firms through the War*. Tech. rep. World Bank.

— (2024). *Ukraine: Firms through the War 2.0*. Tech. rep. World Bank.

Yashiv, Eran (2007). “Labor search and matching in macroeconomics”. In: *European Economic Review* 51.8, pp. 1859–1895.

Online Appendix

A Data Appendix

Dataset Overview

The dataset analyzed in this study comprises job vacancy advertisements collected by Jooble, a job search engine founded in 2006 and active in 67 countries. For Ukraine, Jooble collects postings from 3,490 sources, including 2,570 direct employer feeds and 920 from third-party aggregators. The collected data spans from January 2021 to June 2025 and is provided in JSON format. Each record represents a unique job listing and includes structured fields (e.g., job title, region, salary, number of clicks) and unstructured text (e.g., job descriptions). The initial raw data we have at hand consists of 25 million vacancies.

Key variables include:

- **id**: Unique identifier of the job advertisement.
- **title, description**: Raw job title and full text description.
- **region**: Geographical location of the job.
- **min/max salary, currency, salary rate**: Compensation details (often missing or inconsistent).
- **date created, date expired**: Posting start and end dates.
- **number of clicks**: Proxy for job ad popularity.

Initial analysis shows that job descriptions are predominantly in Ukrainian, though some ads use Russian, English, Polish, or Czech. Salary information is inconsistently reported. Duplicate listings are frequent due to repeated postings or slight changes in metadata. Despite these limitations, the dataset offers rich insight into real-time labor demand trends across regions and occupations.

Pipeline Overview

We developed a modular data processing pipeline using Python and Jupyter Notebooks to clean, enrich, and classify the job ads for labor market analysis. The pipeline includes the following core stages:

1. **Initial Preprocessing**
2. **Candidate Skills Extraction**
3. **ESCO Occupation Classification**
4. **ESCO Skill Enrichment**
5. **Region Enrichment**
6. **Final Data Consolidation**

Each stage was implemented with traceable outputs and a status tracking system to ensure reproducibility and error handling.

Key Processing Steps

Initial Preprocessing (Stage 1) We parse raw JSON files, deduplicate them (based on ID), and clean them. Text fields were sanitized using regex functions to remove dates, salary mentions, URLs, emojis, and special characters. We apply language detection using `fast-langdetect`. We only retain job ads if both title and description were in Ukrainian, Russian, English, Polish, or Czech.

Multi-step Deduplication. Given that Jooble aggregates postings from multiple portals, duplication is a central concern. We address it in two steps. First, we apply content-based deduplication by collapsing postings that share identical job titles, descriptions, salary ranges, and regions, removing repeated appearances of the same underlying vacancy even when different platform-specific identifiers are assigned. Second, we enforce identifier-based deduplication across processing runs to prevent the same vacancy from being reintroduced during incremental updates or partial reprocessing. As a result, each underlying vacancy enters the analysis only once.

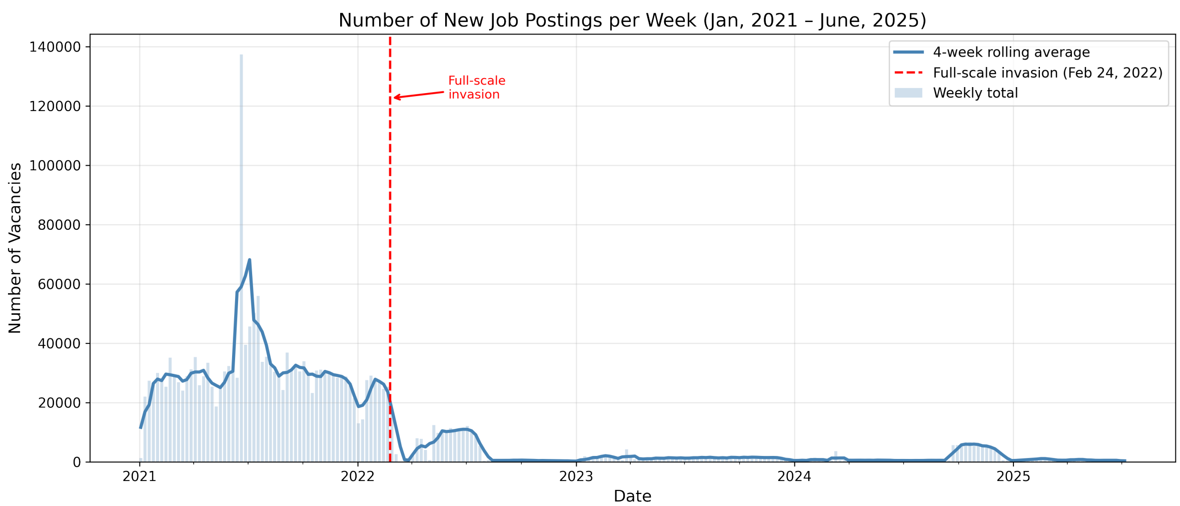
Multi-Region Postings. Identical job advertisements may legitimately appear across multiple regions (e.g., nationwide or multi-location postings). This represents a distinct issue from cross-portal duplication. During intermediate processing stages, such postings are temporarily aggregated to optimize computational efficiency and avoid redundant downstream processing. Importantly, regional information is not discarded. In the final analytical stage, vacancies are re-expanded and assigned to regions, ensuring that regional variation is fully preserved.

Final Vacancy Data. We collapse the data based on the initial date each vacancy was created on the Jooble platform, ensuring that each vacancy appears only once. For the number of clicks—a proxy for job seeker interest or relative attention received by vacancies, which provides insight into potential supply-side preferences—we retain both the average and the maximum number of clicks per vacancy over the total period the vacancy was active. The collapsed data results in 2.2 million vacancies posted between January 2021 and June 2025 (when abstracting from ads that were posted to several regions).

Figure A1 depicts the number of newly posted vacancies captured on the Jooble site between January, 2021 and June, 2025. The plot reveals high weekly fluctuations in postings throughout 2021 and 2022, with a noticeable drop in frequency and volatility in mid-2022. Since mid-2022, the highest number of vacancies was registered from mid-September 2024 to the beginning of December 2024, suggesting increased scraping activity during that period. These developments demonstrate how scraping activity can confound trends in posting activity.

For this reason, the empirical analysis does not rely on raw daily vacancy counts as the main outcome. After deduplication and collapsing repeated postings to their initial posting date, the data are aggregated to weekly and monthly panels. The main dependent variable is constructed as the share of digital skills among all extracted ESCO skills in a given period, which reduces sensitivity to changes in total scraping volume while preserving information on the relative composition of skill demand.

Figure A1: Number of job vacancies posted per week based on Jooble data



Note: Weekly aggregation. Source: Jooble data (Jan, 2021 – June, 2025).

Skill Translation (Interim Step) Since the ESCO framework does not natively provide Russian-language skill labels, we introduced an interim ESCO skill-label translation step before the main skill-extraction stage. All English ESCO skill labels were translated into Russian using the OpenAI GPT-4o API and anchored to their original ESCO concept URIs. The purpose of this step is to expand multilingual matching coverage for Russian-language vacancies without modifying the ESCO taxonomy or introducing new skill categories.

We implement this step as a taxonomy-preserving translation procedure rather than as free-form generation. The relevant ESCO skill and occupation records are translated into Russian to enable matching with Russian-language vacancy texts, but each translated label, alternative label, and description remains linked to the original ESCO concept through its URI or code. Thus, the Russian-language ESCO layer is used only to improve multilingual matching: any matched Russian term is mapped back to the corresponding canonical ESCO skill or occupation. This ensures that the translation step does not introduce new categories or modify the structure of the ESCO taxonomy.

This translation step may introduce noise for rare or highly specialized skill labels. However, the downstream skill-extraction stage relies on semantic similarity rather than exact keyword matching, which reduces sensitivity to minor wording differences in the translated labels. The translated labels are used only for candidate skill retrieval; the occupational classification and final skill enrichment remain anchored to the official ESCO taxonomy.

Candidate Skills Extraction (Stage 2) Using *sentence-transformers* and the *paraphrase-multilingual-mpnet-base-v2* model, we embed sentences from job descriptions and compare them to pre-encoded ESCO skill vectors. Matches are selected via cosine similarity with a threshold of 0.5, chosen to prioritize recall at the candidate extraction stage. This relatively permissive threshold ensures that potentially relevant skills are not prematurely excluded in the initial filtering step. Up to 25 top-matching skills are stored per job ad.

In parallel, we construct a structured `job_types` variable describing the work modality of each posting. Vacancies are classified into four categories: `remote`, `in_office`, `combined`, and `undefined`. This classification combines multilingual regex cues with semantic similarity between vacancy texts and handcrafted reference templates in EN, UA, RU, PL, and CZ, using the multilingual sentence-transformer `paraphrase-multilingual-MiniLM-L12-v2`. Regex rules adjust modality scores and reduce false positives, for example when "remote" appears in an unrelated context. The resulting `job_types` indicator enriches the vacancy-level dataset with information on workplace flexibility while leaving the core skill-extraction procedure unchanged.

Occupation Classification (Stage 3) Stage 3 assigns each posting a provisional ESCO/ISCO-aligned occupation code using a large language model. The model is not used as an open-ended classifier: it operates under a fixed prompt and a predefined JSON output schema requiring three fields: the vacancy identifier, the occupation code, and the occupation title. Where at least five candidate skills have been extracted in Stage 2, the model receives the job title and ranked skill list as structured input. Where fewer candidate skills are available - in the case of short raw vacancy descriptions or vacancies published for marketing purposes - it receives the job title and a truncated vacancy description of up to 1,000 characters. The first-pass classification is performed using *gpt-4o-mini*; low-confidence or likely ambiguous cases are reprocessed using *gpt-4o* - a more powerful model. Outputs that do not conform to the required schema are rejected and reprocessed.

The resulting Stage 3 output is treated as a provisional occupation anchor rather than as the final occupation entry used for skill enrichment. Final occupation resolution occurs in Stage 4, where the provisional code-title pair is verified against the official ESCO taxonomy and resolved to a terminal ESCO occupation entry before the associated skill bundle is retrieved.

A small number of vacancies received non-standard occupation codes from the LLM classification pipeline — for instance, free-text strings such as "data scientist ai nlp" rather than a valid ISCO-08 numeric code. These vacancies account for less than 0.1 % of the total sample and are excluded from all occupation-level analyses. This has no material effect on any reported result.

Stage 3 Prompt Logic and Output Schema The occupation-classification prompt instructs the model to assign each vacancy to a single ESCO/ISCO-aligned occupation category and to return only a structured JSON object. The model is not allowed to introduce new occupation labels or free-text explanations. The required output schema contains three fields:

```
{
  "id": "string",
  "esco_code": "string",
```

```
  "esco_title": "string"
}
```

The field `id` stores the unique vacancy identifier, `esco_code` stores the provisional occupation code, and `esco_title` stores the corresponding occupation title returned by the model. Outputs that do not conform to this schema are rejected and reprocessed. This schema-constrained design ensures that all Stage 3 outputs are machine-readable and suitable for downstream verification in Stage 4.

ESCO Skill Enrichment (Stage 4) In Stage 4, the ESCO verification and skill-enrichment process builds on the provisional occupation output from Stage 3 by resolving each vacancy to a verified terminal ESCO occupation entry. This distinction is important because the Stage 3 output provides a four-digit ESCO/ISCO-aligned occupation anchor, whereas the final skill bundle is retrieved from the terminal ESCO occupation entry in the official taxonomy.

The resolution procedure follows five sequential steps. First, the cleaned job title is matched exactly against ESCO preferred occupation labels. Second, unmatched titles are matched exactly against ESCO alternative labels.[4] Third, fuzzy matching is applied to preferred labels, and fourth, to alternative labels, using a composite similarity score at threshold 0.7. Fifth, for postings where no title-based label match is found, the pipeline applies `esco_code_similarity`, a code-based fallback anchored on the Stage 3 four-digit occupation code. This fallback combines cosine similarity on ESCO occupation embeddings at threshold 0.7 with a hierarchical search through the ESCO taxonomy, traversing downward to the deepest available descendant, then upward to the nearest ancestor if no descendant is found, and finally to the nearest numeric neighbour if both searches fail.

After the five-step verification cascade, only 0.08% of postings remain unclassified. These residual records are excluded from the occupation-linked skill-enrichment step. For all ver-

[4]For each occupation, ESCO provides a preferred ESCO occupation label and alternative occupation labels.

ified postings, we merge the official ESCO skill bundle associated with each matched final occupation entry, ensuring that both explicitly stated and occupation-linked skill requirements are captured in the final analytical dataset.

Once an occupation is identified, its ESCO identifiers and ISCO codes are added, and the vacancy is enriched with all official skills associated with that occupation. This way, even skills not explicitly mentioned in job postings are captured, ensuring a more complete representation of labor market requirements.

Region Enrichment (Stage 5) We parsed original region strings and standardized them into English city, district, region, and country using GPT-4o. Where possible, we added geographic coordinates (latitude, longitude). The enriched region data enables spatial analysis and mapping. Nevertheless, sub-national analysis should be implemented with caution.

Final Data Consolidation (Stage 6) We merged outputs from all stages using the job ID. The final dataset includes fields for cleaned text, detected language, extracted and ESCO skills, occupation codes, and enriched geographic metadata.

Worked Example of Pipeline Transformation

To illustrate how a raw job posting is transformed into an ESCO-linked analytical record, Table A1 presents one representative English-language vacancy and traces it through the main stages of the pipeline, from raw title and detected language to candidate skill extraction, provisional LLM-based occupation classification, final ESCO verification, and occupation-linked skill enrichment.

Table A1: Worked example of vacancy transformation through the data pipeline

Pipeline stage	Field	Value
Raw input	title	Lead Finance Business Controller (Commercial Functions) | Kyiv, Ukraine
Stage 1: Cleaning and language detection	desc_lang	en (English)
	desc_extract	The posting describes a Lead Finance Business Controller vacancy and mentions requirements such as Financial Planning and Analysis (FP&A), financial modeling, budgeting, forecasting, profit and loss (P&L) analysis, key performance indicators (KPIs), business valuation, profitability assessment, commercial finance, data visualization, Microsoft Excel, SQL, and English proficiency.
Stage 2: Skill extraction and work mode	skill_labels (examples)	manage financial aspects of a company, supervise accounting operations, lead managers of company departments, perform financial market business, report on overall management of a business, financial department processes, financial management, analyse business processes, comprehend financial business terminology, ...
	job_type	combined
Stage 3: LLM classification	classified_code	2423
	classified_title	financial manager
Stage 4: ESCO verification and enrichment	esco_title	financial manager
	esco_code	1211.1
	extract_type	preferredLabel
	esco_skills (examples)	financial statements, financial management, financial analysis, enforce financial policies, maintain financial records, ...

Notes: The table reports a shortened vacancy extract and selected skills for readability. Stage 3 outputs are provisional occupation anchors generated under a structured JSON schema. Stage 4 resolves these anchors to verified terminal ESCO occupation entries using the five-step verification cascade described above. The ESCO-enriched skills shown are examples from the official skill bundle linked to the verified ESCO occupation.

Manual Validation and Threshold Sensitivity

To assess the reliability of the occupation-classification and ESCO verification procedure, we conducted a manual validation exercise on a stratified random sample of 200 job postings drawn from the April 2024 dataset. We stratified the sample by two dimensions: the detected language of the vacancy description and the Stage 4 matching method through which the posting was resolved to an ESCO occupation entry. This design ensures that the validation sample reflects both the multilingual structure of the vacancy corpus and the different routes through which records enter the final ESCO-enriched dataset. Table A2 outlines the details.

We evaluate accuracy at successive levels of the ESCO occupational hierarchy by comparing the pipeline-assigned occupation with a manually verified ground-truth occupation. The results show high reliability at broader levels of the hierarchy (Table A3): classification accuracy reaches 91.0% at the one-digit major-group level and 88.0% at the two-digit sub-major-group level. At the four-digit unit-group level, which is the main target of the occupation-classification step and directly relevant for skill enrichment, accuracy reaches 80.0%. Accuracy declines at the most granular full-code level, reflecting the difficulty of assigning noisy multilingual vacancy texts to highly specific ESCO entries, and is lowest for posts in Czech and Polish language (Table A4).

Table A5 further decomposes validation errors by the Stage 4 matching route through which each posting was resolved to an ESCO occupation entry. The results show that label-based matching methods perform substantially better than the final *esco code similarity* fallback at the analytically relevant hierarchy levels. At the four-digit unit-group level, error rates are 6.25 percent for preferred-label matches, 12.50 percent for fuzzy preferred-label matches, 7.14 percent for fuzzy alternative-label matches, and 2.27 percent for alternative-label matches, compared with 48.48 percent for the *esco code similarity* fallback. This pattern is expected because the fallback is applied only after the exact and fuzzy label-based routes fail, and therefore contains more difficult and ambiguous vacancy records. Since the fallback accounts for 66 of the 200 validation cases, the weighted average unit-group error rate remains 20.0 percent, consistent with the overall manual validation accuracy reported above.

This limitation is important for interpreting occupation-linked skill enrichment, but because the empirical analysis is based on aggregated weekly and monthly shares rather than individual vacancy-level classifications, such measurement error is expected primarily to add noise rather than systematically bias the main estimates.

We also conducted a threshold-sensitivity exercise for the Stage 4 `esco_code_similarity` fallback, which is the precision-critical cosine-similarity step in the revised ESCO verification cascade. The exercise compared alternative cosine-similarity thresholds between 0.5 and 0.9 using an independent stratified validation sample. We selected the threshold of 0.7 because it provided the strongest performance at the analytically relevant minor- and unit-group levels while preserving stable accuracy at broader ESCO hierarchy levels. These checks support the use of the revised five-step ESCO verification procedure and provide the empirical basis for the threshold choice used in the final pipeline. Table A6 presents the accuracy rates for different thresholds.

Table A2: Validation sample composition by stratification dimension

Panel A: Posting language				
Language	Population N	Population %	Sample N	Sample %
Russian	6,283	50.0%	99	49.5%
Ukrainian	2,790	22.0%	44	22.0%
Czech	1,833	14.5%	29	14.5%
English	1,317	10.4%	21	10.5%
Polish	456	3.6%	7	3.5%
Total	12,679	100.0%	200	100.0%
Panel B: Stage 4 matching method				
Matching method	Population N	Population %	Sample N	Sample %
`esco_code_similarity`	4,134	33.0%	66	33.0%
Alternative label	2,822	22.0%	44	22.0%
Alternative label fuzzy	2,689	21.0%	42	21.0%
Preferred label	2,011	16.0%	32	16.0%
Preferred label fuzzy	1,023	8.0%	16	8.0%
Total	12,679	100.0%	200	100.0%

Notes: Population refers to classified postings in April 2024. The validation sample was allocated proportionally across posting language and Stage 4 matching method so that it reflects both the multilingual composition of the corpus and the distribution of ESCO verification routes.

Table A3: Manual validation accuracy by ESCO hierarchy level

ESCO level	Description	Accuracy
1	Major group	91.0%
2	Sub-major group	88.0%
3	Minor group	82.5%
4	Unit group	80.0%
6	ESCO sub-unit	58.5%
Full code	Most granular ESCO code	39.0%

Notes: Accuracy is defined as the share of records for which the pipeline-assigned ESCO occupation matches the manually verified ground-truth occupation at the indicated hierarchy level. The validation sample consists of 200 job postings drawn from the April 2024 dataset and stratified by posting language and Stage 4 matching method.

Table A4: Classification error rates by posting language and ESCO hierarchy level

Language	N	Level 1	Level 2	Level 3	Level 4	Level 6	Full code
English	21	4.76%	4.76%	4.76%	9.52%	33.33%	71.43%
Ukrainian	44	6.82%	11.36%	18.18%	20.45%	47.73%	68.18%
Russian	99	5.05%	7.07%	12.12%	15.15%	37.37%	50.51%
Czech	29	27.59%	31.03%	37.93%	37.93%	44.83%	72.41%
Polish	7	14.29%	28.57%	42.86%	42.86%	71.43%	85.71%
Weighted average	200	9.0%	12.0%	17.5%	20.0%	41.5%	61.0%

Notes: Error rate is the share of records for which the pipeline-assigned ESCO occupation does not match the manually verified ground truth at the indicated hierarchy level. Weighted averages use sample cell sizes as weights.

Table A5: Classification error rates by Stage 4 matching method and ESCO hierarchy level

Matching method	N	Level 1	Level 2	Level 3	Level 4	Level 6	Full code
Preferred label	32	3.12%	6.25%	6.25%	6.25%	40.62%	56.25%
Preferred label fuzzy	16	12.50%	12.50%	12.50%	12.50%	12.50%	56.25%
Alternative label fuzzy	42	2.38%	2.38%	4.76%	7.14%	26.19%	40.48%
Alternative label	44	0.00%	0.00%	2.27%	2.27%	22.73%	52.27%
`esco_code_similarity`	66	21.21%	28.79%	42.42%	48.48%	71.21%	83.33%
Weighted average	200	9.0%	12.0%	17.5%	20.0%	41.5%	61.0%

Notes: Matching methods are applied sequentially in Stage 4. The `esco_code_similarity` method is the final fallback step and is therefore expected to contain more difficult cases than exact or fuzzy label matches. Weighted averages use sample cell sizes as weights.

Table A6: Threshold sensitivity of Stage 4 ESCO occupation resolution

ESCO level	th=0.5	th=0.6	th=0.7	th=0.8	th=0.9
Major group	78.2%	87.3%	87.6%	88.2%	88.8%
Sub-major group	71.3%	84.5%	86.9%	87.4%	87.9%
Minor group	56.4%	71.1%	73.0%	72.3%	72.4%
Unit group	44.2%	57.0%	58.4%	55.5%	55.2%
ESCO sub-unit	37.2%	48.6%	50.4%	50.4%	50.0%
Full code	33.0%	44.4%	46.0%	45.4%	44.8%

Notes: The table reports classification accuracy under alternative cosine-similarity thresholds for the Stage 4 `esco_code_similarity` fallback. The threshold-sensitivity exercise uses an independent stratified sample of 200 records from the April 2024 population and is distinct from the final-pipeline manual validation sample reported in Table A3. The adopted threshold is 0.7, which provides the strongest performance at the minor- and unit-group levels while preserving stable accuracy at broader ESCO hierarchy levels.

Output Structure and Use Cases

The enriched dataset is stored in JSON format and includes the following categories:

- **Job metadata**: id, title, description, date created/expired, clicks
- **Cleaned text**: clean title, clean desc, detected languages
- **Extracted skills**: skill URIs and human-readable labels
- **ESCO classification**: occupation codes and titles, ESCO-derived skills
- **Geography**: original and standardized location, coordinates

Verification, Logging, and Operational Stability

Given the computational intensity of several pipeline stages, the workflow incorporates lightweight verification and logging mechanisms to ensure operational stability. Automated checks include row-count validation, uniqueness constraints on identifiers, and summary statistics of missing values to detect unexpected changes in input data across processing stages. Where applicable, hash comparisons are used to monitor potential alterations in intermediate files.

Processing is tracked at the file level with stage-level logging that records execution status and the creation of output artifacts. Because individual stages can be time-consuming, the pipeline is designed to resume from the last successfully completed file in the event of interruption or failure, rather than restarting the entire workflow. This restart mechanism improves robustness to partial runs and facilitates stable, collaborative use across environments.

Table B1: Occupation in the Last 7 Days (2023-2025)

	Frequency	Percent
Managers	2052	10.89
Professionals	6827	36.24
Technicians / associate professionals	1949	10.35
Clerical support workers	1283	6.81
Service and sales workers	2860	15.18
Skilled agricultural workers	490	2.60
Craft and related trades workers	1054	5.60
Plant and machine operators	148	0.79
Elementary occupations	1842	9.78
Other / not classified	332	1.76
Total	18837	100.00

Source: L2UKR 2023-2025.

Percentages are weighted using survey weights.

B Robustness Checks

The Listening to Ukraine (L2UKR) phone surveys are conducted by the World Bank in collaboration with the Kyiv International Institute of Sociology (KIIS) since April 2023. Questions are collected on topics of well-being, damages, incomes, continuity and access to services and utilities, views, among others to assess the conditions of households in Ukraine.

Each month, the L2UKR survey interviews between 1,500 and 2,000 households by phone. These households were originally drawn from a 2021 representative sample of the Ukrainian population. Initially, households are tracked over time until they drop out. When this happens, they are replaced by other randomly assigned households. Due to extremely high attrition, since October 2024, all respondents are contacted through random digit dialing. Random digital dialing makes it possible to cover all parts of Ukraine under Ukrainian cell service, which excludes Luhansk and Crimea. Thus, the survey also includes respondents from regions under active hostilities who remain accessible by phone, although the number of respondents in the Donetsk and Kherson oblasts is small and coverage is limited. When the data are pooled over several rounds, the results can be broken down with reasonable confidence by rural and urban areas, and by regions—with the exception of the regions targeted by Russia, where survey coverage is limited.

Figure B1: Top 50 in-demand skills based on Jooble data (Jan, 2021 - May, 2025)

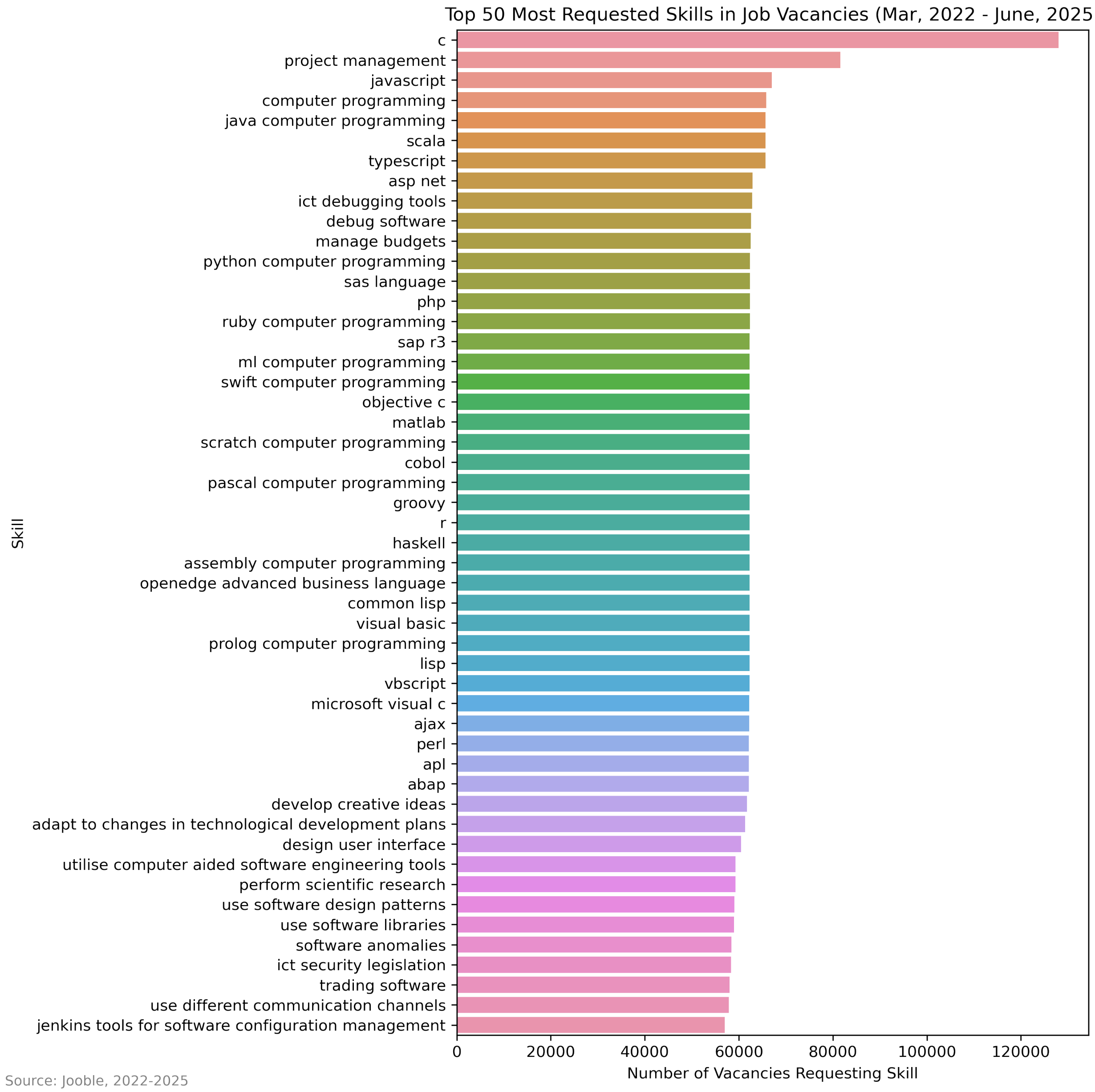


Figure B2: Yearly occupation composition (Jan, 2021 - May, 2025)

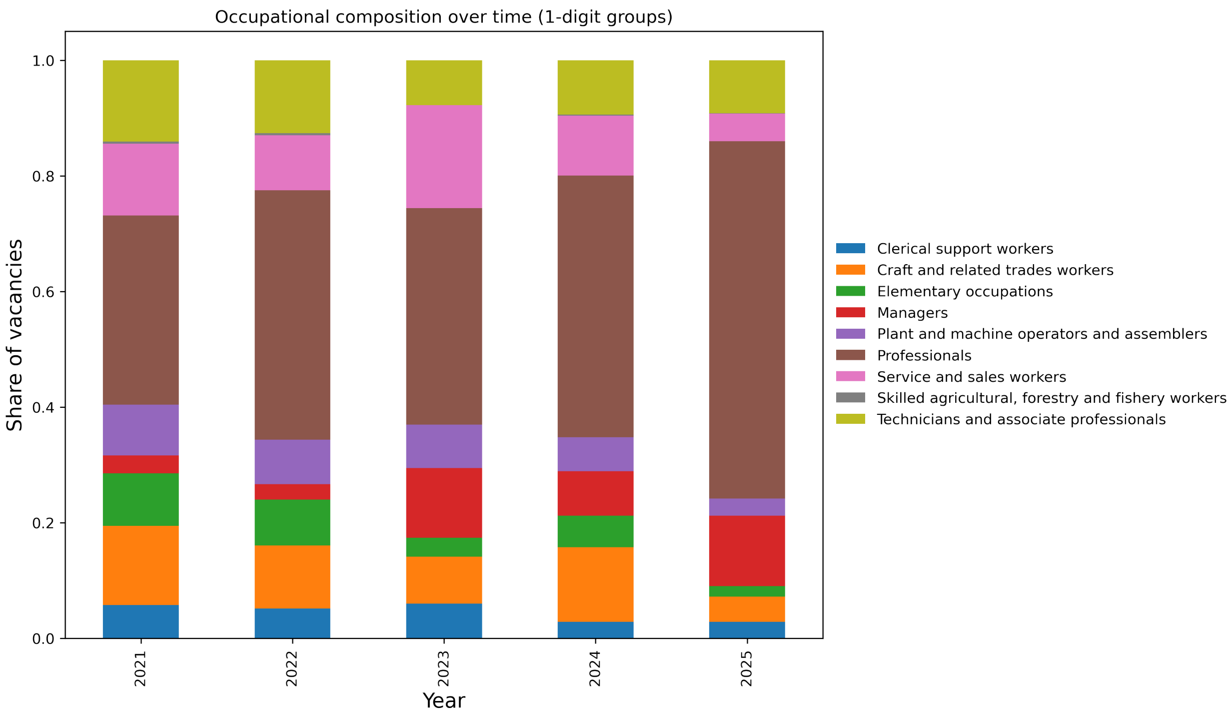
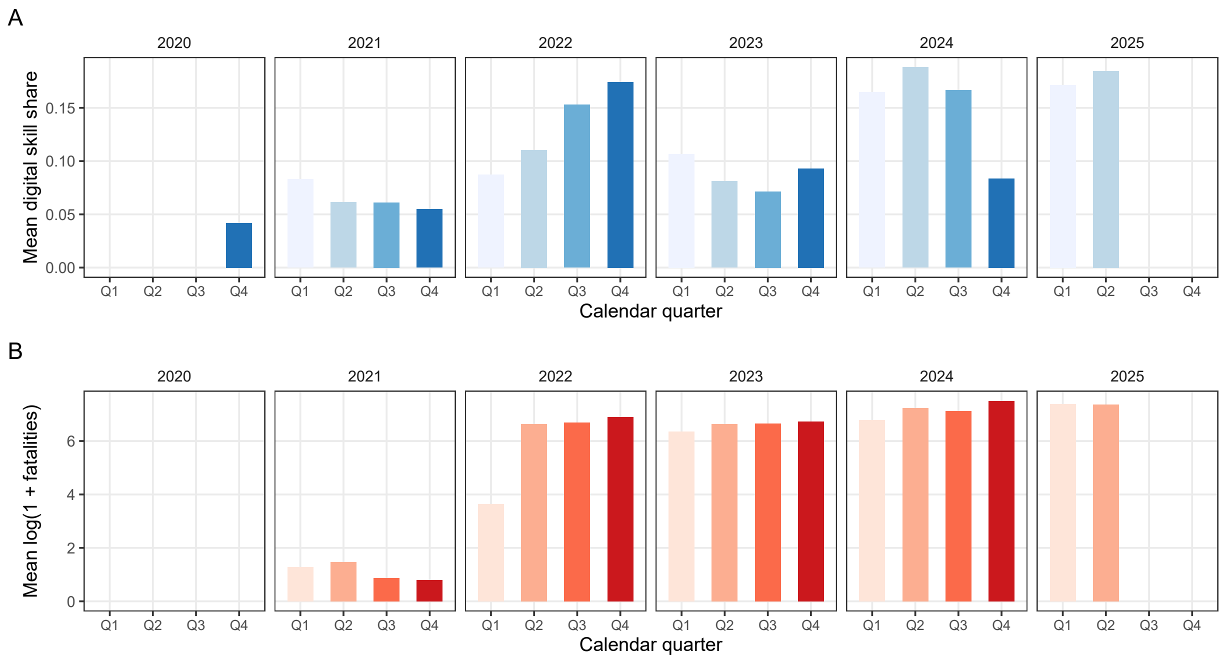
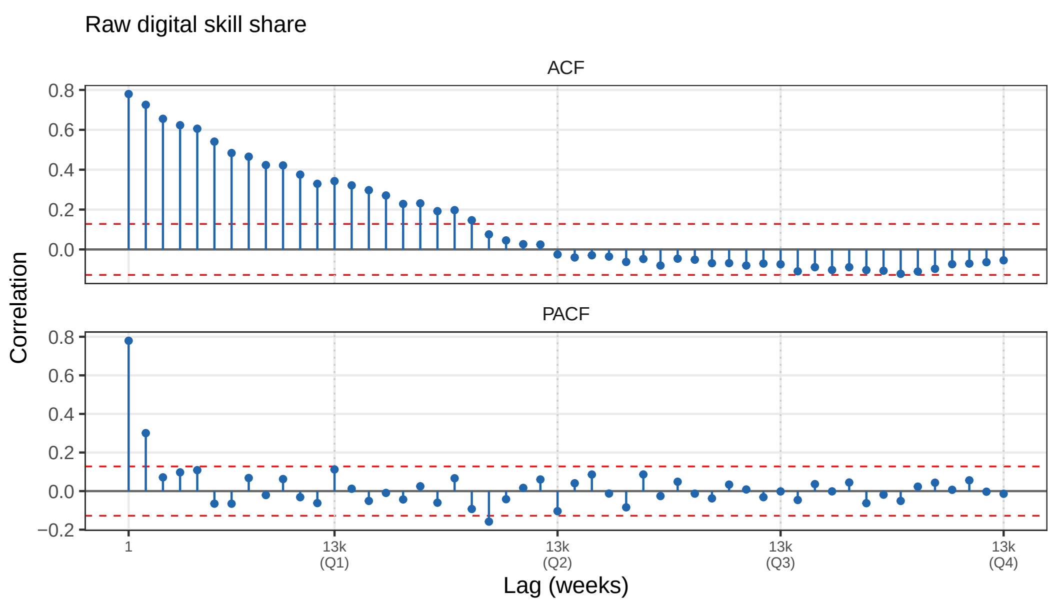


Figure B3: Mean digital skill share (panel A) and mean log-fatalities (panel B) by calendar quarter, 2021-2025.



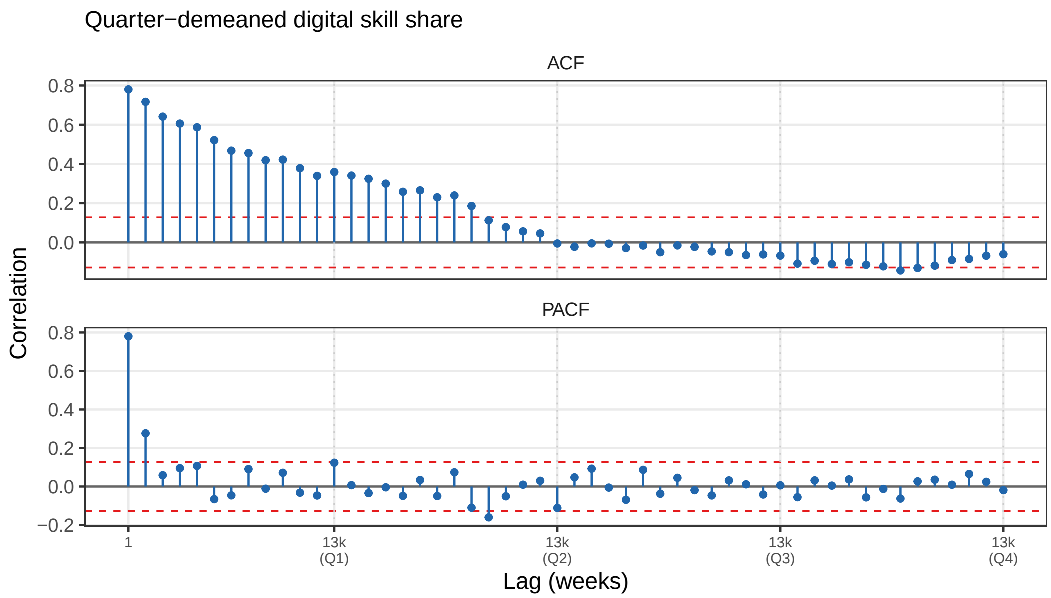
Notes: Bars show within-quarter averages of the weekly series for each year. Systematic within-year variation across Q1–Q4 motivates the inclusion of calendar-quarter fixed effects in all specifications.

Figure B4: Autocorrelation function (ACF) and partial autocorrelation function (PACF) of the raw weekly digital skill share series



Notes: N = 235 observations. Dashed red lines denote the 95% Bartlett confidence band. Dotted vertical lines mark lags that are integer multiples of 13 weeks (one calendar quarter). Significant positive autocorrelation at quarterly multiples indicates quarterly periodicity in the series.

Figure B5: ACF and PACF of the digital skill share after removing calendar-quarter group means (quarter-demeaned residuals).



Peaks at quarterly lags visible in Figure B4 attenuate markedly after demeaning, confirming that calendar-quarter fixed effects absorb the seasonal signal.

Table B2: Seasonality Diagnostics: Tests for Quarterly Patterns

	Series	Statistic	p-value
F-test: joint significance of Q2, Q3, Q4 dummies	Digital skill share	$F(3, N) = 2.00$	0.115
F-test: joint significance of Q2, Q3, Q4 dummies	Log(1 + fatalities)	$F(3, N) = 1.04$	0.376
Friedman (H_0: no consistent seasonal profile across years)	Digital skill share	$\chi^2 = 0.60$	0.896
Kruskal–Wallis (H_0: equal distribution across quarters)	Digital skill share	$H = 9.95$	0.019**
Kruskal–Wallis (H_0: equal distribution across quarters)	Log(1 + fatalities)	$H = 4.94$	0.176

The Kruskal–Wallis test is a non-parametric test of the null hypothesis that the distribution of the series is identical across the four calendar quarters. The F-test is the joint significance test of Q2, Q3, Q4 dummies in OLS (HC3 robust standard errors; Q1 is the omitted category). The Friedman test evaluates whether the within-year seasonal profile (quarterly means) is consistent across years; it requires a balanced year × quarter panel and is reported for the digital skill share only. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B3: Monthly Aggregation Robustness: Digital Skill Demand and Fatalities

	(1)	(2)	(3)	(4)
Log(Fatalities)	0.011***	0.011***	0.014***	0.011***
	(0.002)	(0.002)	(0.004)	(0.002)
Intercept	0.033***	0.037**	0.031**	0.055***
	(0.009)	(0.017)	(0.014)	(0.019)
Num.Obs.	54	54	54	54
R2	0.303	0.331	0.533	0.385
R2 Adj.	0.290	0.277	0.450	0.205

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: average digital skill share (monthly frequency, vacancy-count-weighted mean of weekly values). Log(Fatalities) is log(1 + monthly sum of ACLED fatalities). Columns mirror the main weekly OLS table (Table 2): (1) no fixed effects, HC3 standard errors; (2) quarter fixed effects, HAC SE (Newey-West, 3 lags); (3) quarter and year fixed effects, HAC SE; (4) month-of-year fixed effects (12 indicators), HAC SE. 3-lag Newey-West is the monthly analogue of the 4-lag choice in the weekly models, both spanning approximately one quarter.

Figure B6: CUSUM test for parameter stability (OLS residuals)

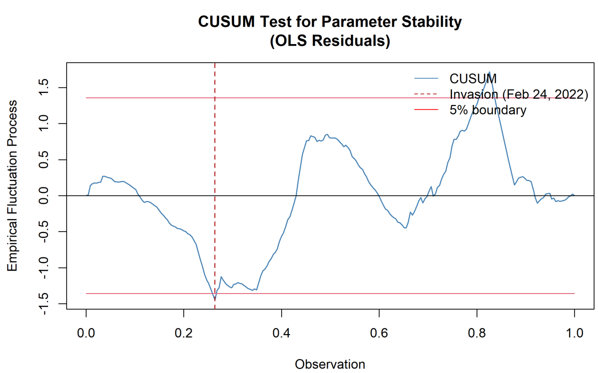
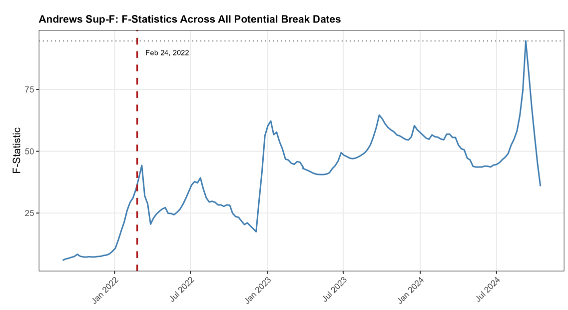


Figure B7: Andrews Sup-F F-statistics across all candidate break dates



Notes: Based on Jooble data (Jan, 2021-June, 2025).

Table B4: Monthly Aggregation Robustness: Digital Skill Demand and Events

	(1)	(2)	(3)	(4)
Log(Events)	0.033***	0.033***	0.050***	0.034***
	(0.005)	(0.007)	(0.011)	(0.007)
Intercept	−0.153***	−0.149***	−0.254***	−0.140**
	(0.038)	(0.051)	(0.071)	(0.053)
Num.Obs.	54	54	54	54
R2	0.292	0.331	0.562	0.383
R2 Adj.	0.279	0.276	0.484	0.202

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: average digital skill share (monthly frequency, vacancy-count-weighted mean of weekly values). Log(Events) is log(1 + monthly sum of ACLED events). Columns mirror the main weekly OLS table (Table 2): (1) no fixed effects, HC3 standard errors; (2) quarter fixed effects, HAC SE (Newey-West, 3 lags); (3) quarter and year fixed effects, HAC SE; (4) month-of-year fixed effects (12 indicators), HAC SE. 3-lag Newey-West is the monthly analogue of the 4-lag choice in the weekly models, both spanning approximately one quarter.

Table B5: Structural Break Test Statistics

Test	Statistic	p-value	H0 rejected
Andrews Sup-F (unknown break date)	94.694	<0.001	Yes
Chow test (Feb 24, 2022)	7.382	<0.001	Yes
CUSUM (OLS)	1.721	0.00534	Yes

Null hypothesis (H_0): parameter stability (no structural break). Sup-F and Chow tests use an F-distribution; CUSUM uses the Kolmogorov–Smirnov boundary. All tests estimated on `avg_digital_skill_share` regressed on a linear time trend, ChatGPT rollout dummy, and quarter fixed effects. Break date for Chow test: February 24, 2022.

Table B6: Digital Skill Share Regressions: Continuous vs. Structural Break Specifications

	(1) Continuous	(2) Inv. Dummy	(3) ITS	(4) ITS+Ctrls	(5) ITS+Cont.
Log(Fatalities)	0.012***				−0.004
	(0.001)				(0.004)
Post-2022 (0/1)		0.073***	0.064***	0.056***	0.078***
		(0.008)	(0.017)	(0.019)	(0.023)
Linear Time Trend			0.000	0.000	0.000
			(0.000)	(0.000)	(0.000)
Time Since February 2022 (slope shift)			0.001**	0.001***	0.001***
			(0.000)	(0.000)	(0.000)
Vacancy Count				0.000*	0.000*
				(0.000)	(0.000)
ChatGPT Rollout (0/1)				−0.068***	−0.071***
				(0.021)	(0.022)
Military Skill Share				0.515	0.517
				(0.711)	(0.710)
Intercept	0.060***	0.073***	0.084***	0.112***	0.116***
	(0.009)	(0.008)	(0.008)	(0.017)	(0.020)
Num.Obs.	235	235	235	235	235
R2	0.328	0.322	0.356	0.464	0.465
R2 Adj.	0.316	0.310	0.339	0.442	0.441

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included in all models but not shown. Dependent variable: `avg_digital_skill_share` (weekly). Full-scale targeted date: Feb 24, 2022. Column labels: (1) continuous log(fatalities), no break; (2) post-2022 dummy only; (3) ITS with level and slope shift at Feb 24, 2022; (4) ITS with full controls; (5) ITS and continuous regressor simultaneously.

Table B7: Unit Root and Stationarity Tests

	ADF (H_0: unit root)		PP (H_0: unit root)		KPSS Level (H_0: stationary)		KPSS Trend (H_0: stationary)	
Series	ADF stat	ADF p	PP stat	PP p	KPSS (Level) stat	KPSS (Level) p	KPSS (Trend) stat	KPSS (Trend) p
Avg. Digital Skill Share	-3.312	0.0701	-6.070	0.01	1.622	0.01	0.126	0.0863
Log(Fatalities)	-1.647	0.724	-2.664	0.297	3.329	0.01	0.694	0.01

ADF = Augmented Dickey–Fuller test (lag selected by AIC). PP = Phillips–Perron test (constant + trend specification). KPSS = Kwiatkowski–Phillips–Schmidt–Shin test. For ADF and PP, rejection of H_0 indicates stationarity. For KPSS, rejection of H_0 indicates nonstationarity. p-values for ADF and KPSS are interpolated from standard tables; reported as <0.01 when below that threshold.

Table B8: Engle–Granger Cointegration Tests (ADF on Long-Run Residuals)

Long-run specification	ADF stat	p-value	Cointegration
Outcome $\sim$ Log(Fatalities) + trend	-3.365	0.0613	Yes (10%)
Outcome $\sim$ ITS vars + trend	-3.375	0.0596	Yes (10%)
Outcome $\sim$ ITS + all controls + trend	-3.934	0.0130	Yes (5%)

Engle–Granger two-step procedure: ADF test applied to OLS residuals from the long-run (levels) regression. H_0: residuals contain a unit root (no cointegration). Rejection indicates cointegration: the levels regression captures a genuine long-run equilibrium, not a spurious correlation. Lag length selected automatically by AIC. Note: p-values are based on standard ADF critical values; correct Engle–Granger critical values for residual-based tests are more negative, so reported p-values are conservative. ‘Yes (10

We estimate an Error Correction Models (ECM) following the following specification:

$$Deltay_t = alpha + beta_1 Deltax_t + gammahatu_{t-1} + varepsilon_t \tag{8}$$

where $hatu_{t-1}$ is the lagged residual from the corresponding long-run (levels) regression. A negative and significant ECM term $gamma$ indicates cointegration: the long-run levels relationship is genuine, not spurious.

Table B9: Error Correction Models: Short-Run Dynamics and Cointegration

	(1) Fatalities	(2) ITS	(3) Full
ΔLog(Fatalities)	-0.002		-0.003
	(0.003)		(0.003)
ΔPost-2022 (impulse)		0.003	-0.002
		(0.004)	(0.004)
ΔTime Since February 2022		0.001	0.000
		(0.004)	(0.004)
ΔVacancy Count			0.000***
			(0.000)
ΔChatGPT Rollout			0.039***
			(0.006)
ΔMilitary Skill Share			0.551
			(0.381)
ECM term $\hat{u}_{t-1}$	-0.333***	-0.337***	-0.425***
	(0.071)	(0.073)	(0.083)
Intercept	0.006	0.005	0.007
	(0.004)	(0.004)	(0.004)
Num.Obs.	234	234	234
R2	0.172	0.172	0.246
R2 Adj.	0.154	0.151	0.213

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Error Correction Models (ECM): $\Delta y_t = \alpha + \beta_1 \Delta x_t + \gamma \hat{u}_{t-1} + \varepsilon_t$, where $\hat{u}_{t-1}$ is the lagged residual from the corresponding long-run (levels) regression. A negative and significant ECM term γ indicates cointegration: the long-run levels relationship is genuine, not spurious. HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included but not shown.

Table B10: Detrended Regressions: Digital Skill Share (Trend-Stationary Approach)

	(1) Fatalities	(2) ITS	(3) Full
Log(Fatalities) [detrended]	0.008***		0.000
	(0.003)		(0.005)
Post-2022		0.038***	0.034
		(0.014)	(0.022)
Time Since February 2022		0.000	0.000
		(0.000)	(0.000)
Vacancy Count [detrended]			0.000*
			(0.000)
ChatGPT Rollout			−0.067***
			(0.022)
Military Skill Share [detrended]			0.723
			(0.730)
Intercept	0.010	−0.004	0.004
	(0.007)	(0.009)	(0.009)
Num.Obs.	235	235	235
R2	0.111	0.105	0.259
R2 Adj.	0.096	0.085	0.230

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Detrended specifications: the dependent variable and continuous regressors are residualised on a linear time trend before estimation, removing the deterministic trend component common to both series. Step/ramp dummies (Post-2022, Time Since February 2022, ChatGPT Rollout) are included in levels as they represent structural breaks, not trends. HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included but not shown.

Table B11: Digital Posting Count: Specifications Across Models

	(1) Fatalities	(2) ITS	(3) Full
Log(Fatalities)	−0.438***		−0.130*
	(0.023)		(0.067)
Post-2022		−2.557***	−0.615
		(0.302)	(0.626)
Time Since February 2022		−0.001	0.010**
		(0.003)	(0.005)
ChatGPT Rollout			−0.760**
			(0.334)
Military Skill Share			−14.474
			(10.640)
Time Trend			−0.003
			(0.004)
Vacancy Count			0.000**
			(0.000)
Intercept	10.395***	9.970***	9.351***
	(0.155)	(0.123)	(0.536)
Num.Obs.	235	235	235
R2	0.719	0.753	0.814
R2 Adj.	0.714	0.748	0.806

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: log(1 + total digital skill postings). HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included but not shown.

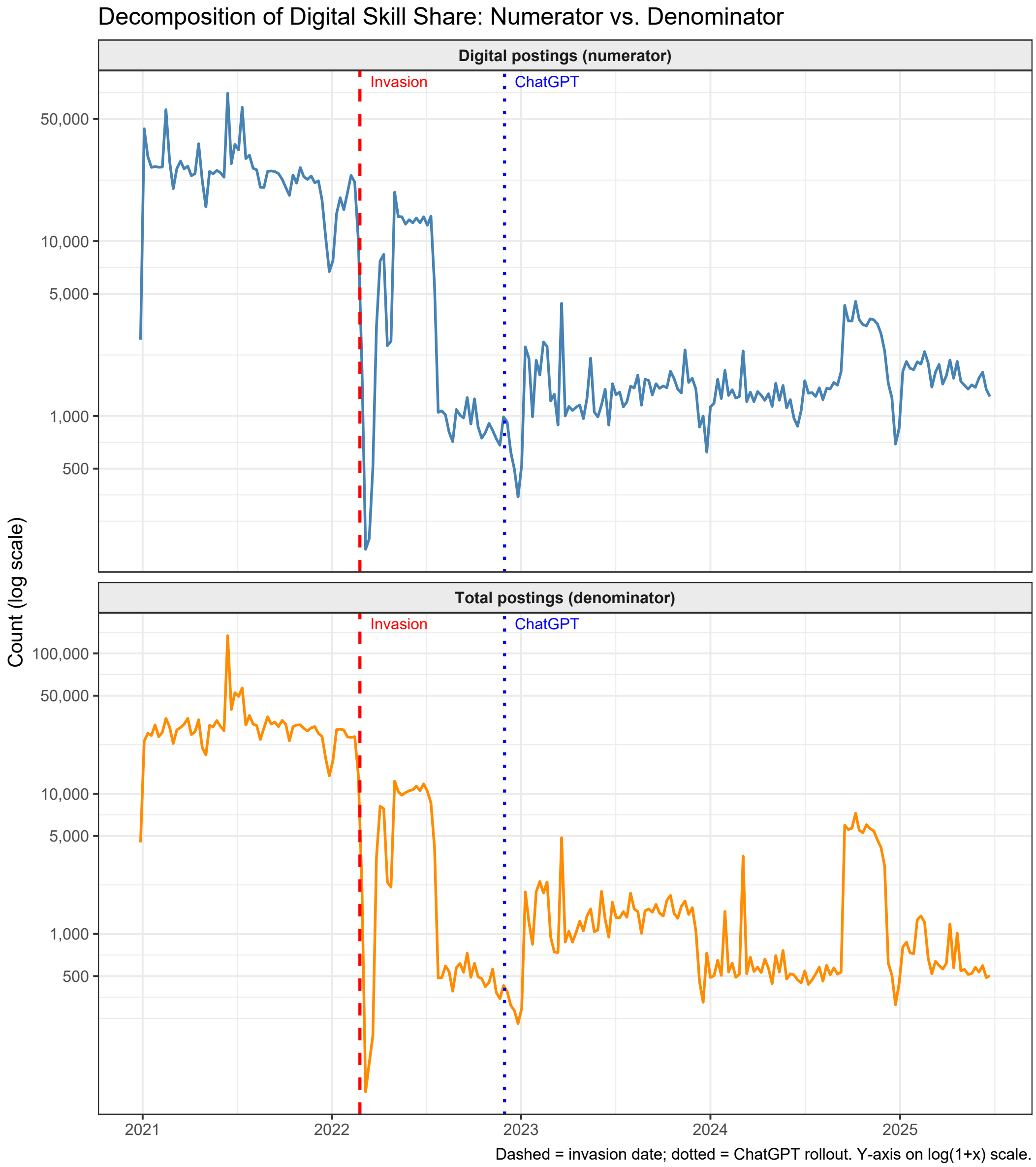
Table B12: Decomposition of Digital Skill Share: Numerator and Denominator Regressions

	Digital (numerator)	Total (denominator)
Log(Fatalities)	−0.130*	−0.055
	(0.067)	(0.101)
Post-2022	−0.615	−2.651***
	(0.626)	(0.692)
Time Since February 2022	0.010**	−0.001
	(0.005)	(0.007)
ChatGPT Rollout	−0.760**	−0.207
	(0.334)	(0.463)
Military Skill Share	−14.474	−13.685
	(10.640)	(15.906)
Time Trend	−0.003	0.000
	(0.004)	(0.005)
Vacancy Count	0.000**	
	(0.000)	
Intercept	9.351***	10.254***
	(0.536)	(0.251)
Num.Obs.	235	235
R2	0.814	0.748
R2 Adj.	0.806	0.737

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variables: log(1 + total digital skill postings) [numerator] and log(1 + total vacancy count) [denominator]. These models decompose the digital skill share into its numerator and denominator to distinguish genuine upskilling from a collapse in overall labour demand. HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included in all models but not shown.

Figure B8: Decomposition of Digital Skill Share: Numerator and Denominator over Time.



Notes: The upper panel plots weekly total digital skill postings (numerator); the lower panel plots total weekly vacancy postings (denominator). Both series are plotted on a $\log(1+x)$ scale, consistent with the regression specifications in Tables B11 and B12. Dashed red line = targeted date (24 February 2022); dotted blue line = ChatGPT rollout (30 November 2022).

Table B13: First Stage: Remote Work Share and Attack Intensity

	(1) Fatalities	(2) Fatalities + Ctrls	(3) Events	(4) Events + Ctrls
Log(Fatalities)	0.041***	0.027***		
	(0.004)	(0.007)		
Log(Events)			0.118***	0.064**
			(0.011)	(0.025)
Vacancy Count		0.000*		0.000*
		(0.000)		(0.000)
ChatGPT Rollout		−0.042		−0.048
		(0.051)		(0.055)
AI Wave 2		0.081		0.091*
		(0.051)		(0.053)
Intercept	0.050**	0.131***	−0.487***	−0.133
	(0.023)	(0.042)	(0.064)	(0.163)
Num.Obs.	235	235	235	235
R2	0.432	0.461	0.394	0.438
R2 Adj.	0.422	0.444	0.384	0.421

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: weekly remote work share (share of job postings explicitly mentioning remote or distance work). Columns (1) and (3): baseline specification with quarter fixed effects only. Columns (2) and (4): full controls — vacancy count, ChatGPT rollout, and AI Wave 2. Columns (1)–(2) use Log(Fatalities); columns (3)–(4) use Log(Events). A significant positive coefficient confirms that remote work adoption is itself attack-induced, justifying the mediator classification. HAC standard errors (Newey-West, 4 lags) in parentheses. Quarter fixed effects included but not shown.